Data Review

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Intro here…

The reason I choose this topic was because previously I was helping the Allentown Police Department by creating a perception survey with two other people. We went and looked in to the backgournd of Allentown and other cities that closely resemble Allentown. Once we completed our survey I thought that a good topic/idea to research next was gender differences in police officers and crimes. I wish to see if there was some kind of graph that could help me visualize how crimes and police officer gender is played out. This was a topic that I always found intersting because I wished to know how a male dominated job effects something else, which brought me to police officers and crimes.

Talk about things…

Histograms are here…

Below is the data I collected and normalized and some histograms graphed out. This will help me see the amount in each group.

```
GenderResearch<-read.file("/home/an248493/Final2.csv")
```

```
## Reading data with read.csv()
```

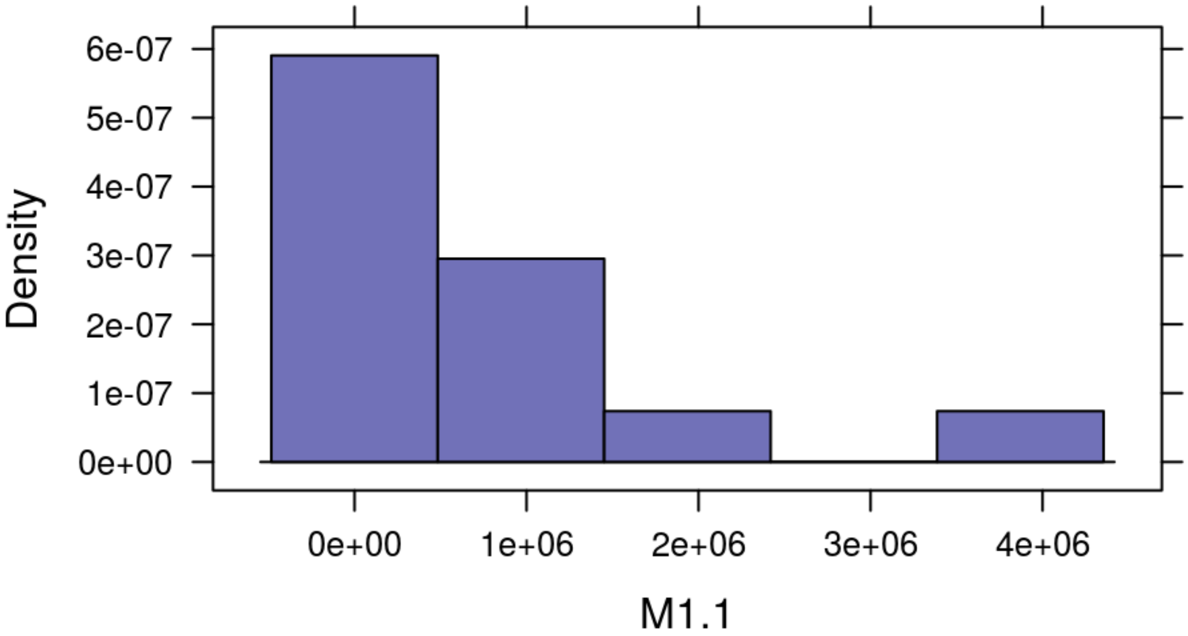
```
GenderResearch
```

```
##                  X    M1.1    M1.2        M1.3         M1.4      M1.5
## 1   San Francisco   408462  396773 0.003580610  0.000056200  2.38e-06
## 2      Northampton   12312   16237 0.004055706 -0.000019600 -1.96e-05
## 3        Baltimore  292249  328712 0.005924691  0.000231534 -3.80e-07
## 4      Bloomington   40161   43162 0.001215709  0.000028400  4.98e-05
## 5          Detroit  337679  376098 0.004432267  0.000075200 -3.03e-07
## 6     Philadelphia  719813  806193 0.004876967  0.000316352  1.67e-05
## 7          Chicago 1308072 1387526 0.005694139  0.000199140  7.01e-06
## 8      Los Angeles 1889064 1903557 0.003080120  0.000061500  1.61e-06
## 9          Memphis  307019  339870 0.004225503  0.000177459  1.33e-05
## 10   New York City 3883188 4291945 0.006300444  0.000136866  2.39e-06
## 11          Dallas  598962  598854 0.003415751  0.000005010  1.67e-05
## 12          Boston  295951  321643 0.005171140  0.000085000  1.35e-05
## 13         Houston 1053517 1045934 0.003539804  0.000067300  2.81e-06
## 14 Las Vegas Metro  294100  289656 0.006771982  0.000138626  2.71e-05
## 15                      NA      NA          NA           NA        NA
## 16                      NA      NA          NA           NA        NA
##           M1.6        M1.7        M1.8        M1.9       M1.10
## 1   0.000046500 0.000668713 0.002012034 0.000714861 0.000561180
## 2  -0.000019600 0.000406108 0.001515517 0.000506963 0.002711521
```

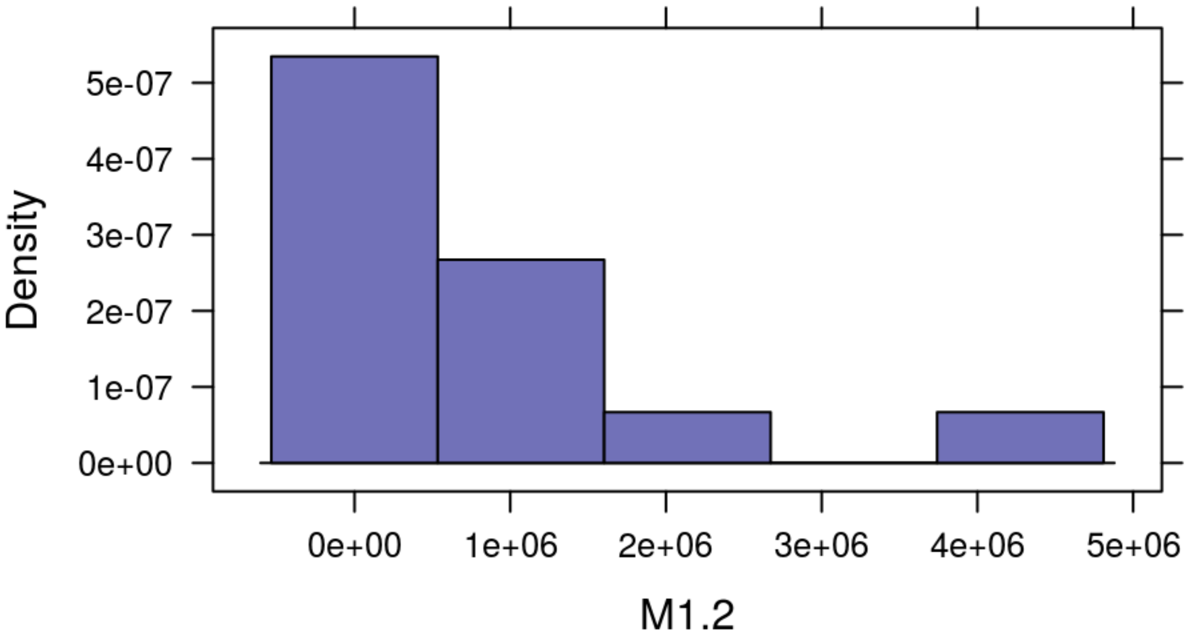
```
## 3   0.000359283 0.002161327 0.001998494 0.003779357 0.002376687
## 4   0.000024900 0.000323697 0.001397188 0.001390263 0.002822227
## 5   0.000133263 0.001336512 0.001993624 0.002356504 0.001049145
## 6   0.000441781 0.003179796 0.002749430 0.001978987 0.002993893
## 7   0.000285152 0.001563648 0.001617221 0.001479242 0.004425441
## 8   0.000104288 0.000892785 0.002711122 0.000877836 0.000697762
## 9   0.000323085 0.002015194 0.002499093 0.004052352 0.002881007
## 10  0.000234098 0.002607259 0.002981240 0.001086258 0.002120680
## 11  0.000153599 0.000489166 0.000923153 0.001013401 0.001518798
## 12  0.000168947 0.001625518 0.004758666 0.001372638 0.002486951
## 13  0.000158517 0.000813727 0.000839897 0.001022496 0.001912779
## 14  0.000380823 0.002444618 0.004088367 0.002489134 0.004440234
## 15           NA          NA          NA          NA          NA
## 16           NA          NA          NA          NA          NA
##          M1.11       M1.12       M1.13        M1.14        M1.15
## 1   0.000230055  4.1400e-05 0.001600447  0.000074800  0.000087100
## 2  -0.000061600 -1.9600e-05 0.004989242 -0.000123175 -0.000061600
## 3   0.001657589  1.4028e-04 0.011637233  0.000210986 -0.000009520
## 4  -0.000068400  5.1500e-05 0.003114355  0.000008660  0.000103062
## 5   0.001283334  9.0100e-05 0.004008707  0.000083900  0.000141255
## 6   0.001026160  5.1600e-05 0.004109454  0.000101415  0.000315906
## 7   0.001340519  1.8000e-05 0.007129246  0.000031500  0.000082000
## 8   0.000241342  1.4900e-05 0.001252753  0.000179982  0.000099400
## 9   0.000511139  9.4300e-05 0.008371858  0.000346794  0.000494310
## 10  0.000745215  2.9600e-05 0.004523198  0.001630570  0.003898319
## 11  0.000285482 -3.0100e-10 0.004435726  0.000218682  0.000125204
## 12  0.000185971  6.7600e-06 0.004044314  0.000018600  0.000149063
## 13  0.000014200  1.2300e-05 0.005128819  0.000184502  0.001072363
## 14  0.000673458  2.3800e-05 0.011738076  0.000436411  0.001131836
## 15           NA          NA          NA           NA           NA
## 16           NA          NA          NA           NA           NA
##         M1.16         M1.17       M1.18       M1.19        M1.20
## 1  -2.16e-07  2.987490e-04 0.000508141 0.000175910  0.000126501
## 2  -1.96e-05  1.820770e-04 0.000484643 0.000081200 -0.000019600
## 3  -3.80e-07 -3.800000e-07 0.000946202 0.002953652 -0.001292274
## 4  -1.73e-06  7.521870e-04 0.000983210 0.000151130 -0.000001730
## 5  -1.03e-05  4.979440e-04 0.000408449 0.003917763  0.000000908
## 6   3.83e-05  6.150000e-05 0.000488520 0.001626862 -0.000807250
## 7   7.64e-07  1.030000e-06 0.001657038 0.002744294 -0.000812212
## 8   2.27e-06  2.677150e-04 0.000765395 0.001098086 -0.000350501
## 9   2.11e-05  3.269600e-04 0.000893338 0.002798055  0.000442187
## 10  1.68e-05  5.378010e-04 0.001721009 0.001957241 -0.000369652
## 11  1.67e-06  2.500000e-05 0.000220367 0.001172010 -0.001451228
## 12 -2.84e-06  7.268610e-04 0.000847283 0.001071253 -0.000213075
## 13 -8.62e-06  6.880000e-09 0.000551120 0.001299078 -0.001540458
## 14  3.70e-05  1.841886e-03 0.002855303 0.002538084 -0.008318148
## 15        NA            NA          NA          NA           NA
## 16        NA            NA          NA          NA           NA
##          M1.21       M1.22        M1.23         M1.24        M1.25
## 1  0.000183255 0.002393089  1.178073e-03  0.0012248090  0.000004900
```

```
## 2  0.000182077 0.002921290  5.881850e-04  0.0023331060 -0.000019600
## 3  0.000345964 0.044760613  1.367715e-02  0.0310834610  0.000429615
## 4  0.000750456 0.005640282  8.372740e-04  0.0048030080 -0.000001730
## 5  0.000156954 0.007538862  1.619293e-03  0.0059106850  0.000005920
## 6  0.000678797 0.019223900  9.427104e-03  0.0097967950  0.000102408
## 7  0.000377941 0.022083700 -4.380000e-08 -0.0000000438  0.001859969
## 8  0.000548859 0.003843449  1.019464e-03  0.0028239850  0.000107012
## 9  0.000429517 0.015456361  4.691429e-03  0.0106881300  0.000384971
## 10 0.001329357 0.022262072  5.993811e-03  0.0162682610  0.001083118
## 11 0.000293838 0.006549362  7.796570e-04  0.0057697040  0.000005010
## 12 0.000405332 0.006986258  4.781913e-03  0.0022043440  0.000013500
## 13 0.000383216 0.009423177  6.880000e-09  0.0094231770  0.000067400
## 14 0.002501725 0.021534527  5.018165e-03  0.0165163620  0.000016800
## 15          NA          NA            NA            NA           NA
## 16          NA          NA            NA            NA           NA
##           M1.26       M1.27       M1.28         M1.29       M1.30
## 1   7.21000e-08 0.000669859 0.000106639  0.0028565140 0.000094000
## 2  -1.96000e-05 0.006041605 0.001028188  0.0107506010 0.001795764
## 3   8.21000e-05 0.000239510 0.000117852 -0.0000003800 0.002091952
## 4  -1.50061e-04 0.002197188 0.003255504  0.0092033440 0.000294673
## 5   2.22000e-05 0.001356923 0.000005920  0.0000062300 0.000741893
## 6  -1.99000e-05 0.004339622 0.000527221  0.0003620490 0.003182428
## 7  -1.90000e-05 0.001979624 0.000262679 -0.0000000438 0.006266655
## 8  -2.10000e-06 0.002635835 0.001509324  0.0000026500 0.000054200
## 9   7.46000e-06 0.002609397 0.004329790  0.0035045750 0.007903803
## 10  2.49000e-06 0.001008229 0.032554339 -0.0000000245 0.009702215
## 11  7.51000e-05 0.002006708 0.000018400  0.0115362050 0.000762965
## 12  3.38000e-06 0.000460864 0.000023900 -0.0000002700 0.001596296
## 13  7.12000e-05 0.003954741 0.001591397  0.0084903960 0.002009112
## 14 -1.88401e-04 0.012362682 0.008582208  0.0000034000 0.002099832
## 15           NA          NA          NA            NA          NA
## 16           NA          NA          NA            NA          NA
##            M1.31       M1.32         M1.33       M1.34       M1.35 X.1
## 1   1.796530e-04 0.011657421  0.0000000721 0.002783500 0.001547500  NA
## 2  -1.960000e-05 0.016885003 -0.0000196000 0.001921625 0.003156896  NA
## 3   2.098650e-04 0.035051307  0.0004626830 0.004750638 0.007953913  NA
## 4  -1.730000e-06 0.013408653  0.0000405000 0.001774147 0.004195584  NA
## 5   6.400000e-05 0.006036210  0.0000560000 0.003538643 0.004779035  NA
## 6   4.970040e-04 0.015263150  0.0124613330 0.006687359 0.006050591  NA
## 7  -4.380000e-08 0.008369671 -0.0000000438 0.003665161 0.007263180  NA
## 8   5.640000e-05 0.013434843  0.0004180820 0.003769719 0.001831803  NA
## 9   7.140000e-06 0.024492150  0.0004658190 0.005014832 0.007538846  NA
## 10  1.701470e-04 0.069597529 -0.0000000245 0.005959463 0.003981707  NA
## 11  3.010000e-10 0.012974734  0.0000117000 0.001570926 0.002817680  NA
## 12 -2.700000e-07 0.002127070 -0.0000002700 0.006638144 0.004052317  NA
## 13  2.077440e-04 0.012102127  0.0003012190 0.001879459 0.002961812  NA
## 14  3.112599e-03 0.044162321  0.0013499800 0.007052433 0.007626627  NA
## 15            NA          NA            NA          NA          NA  NA
## 16            NA          NA            NA          NA          NA  NA
```

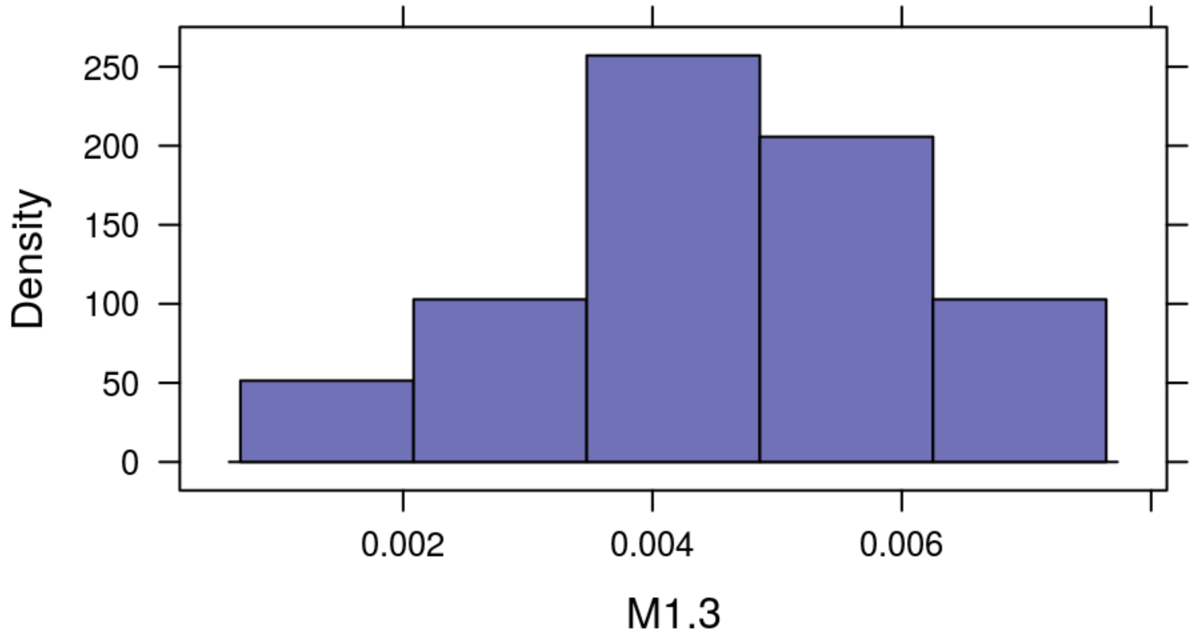
```
histogram(~M1.1,data=GenderResearch)
```



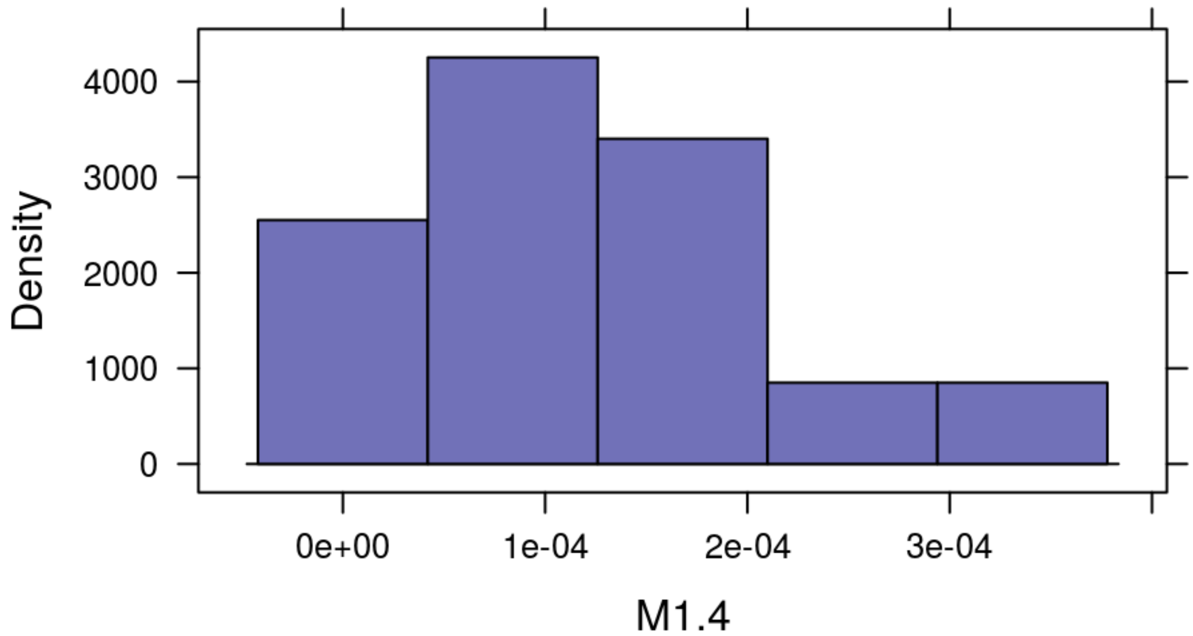

```
histogram(~M1.2,data=GenderResearch)
```



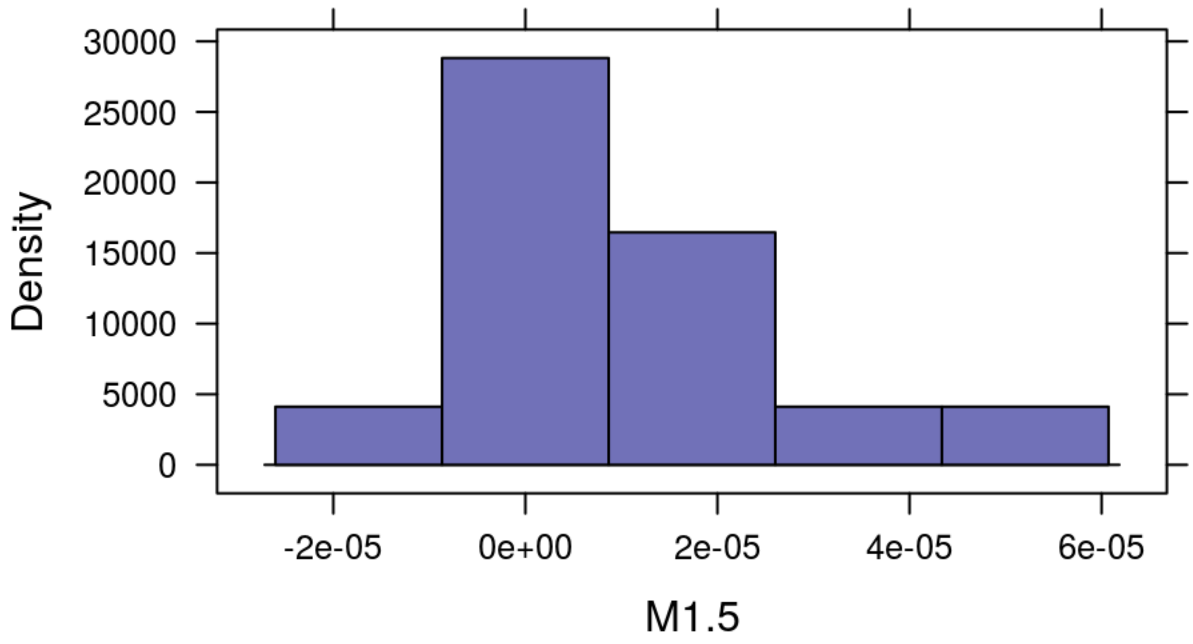

```
histogram(~M1.3,data=GenderResearch)
```



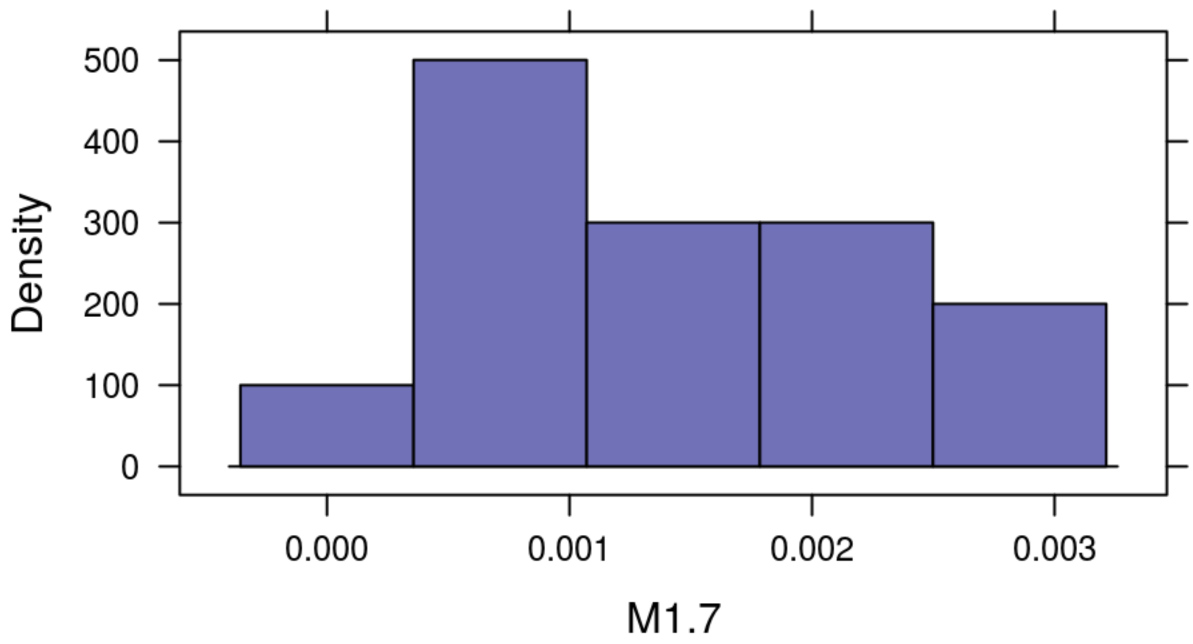

```
histogram(~M1.4,data=GenderResearch)
```



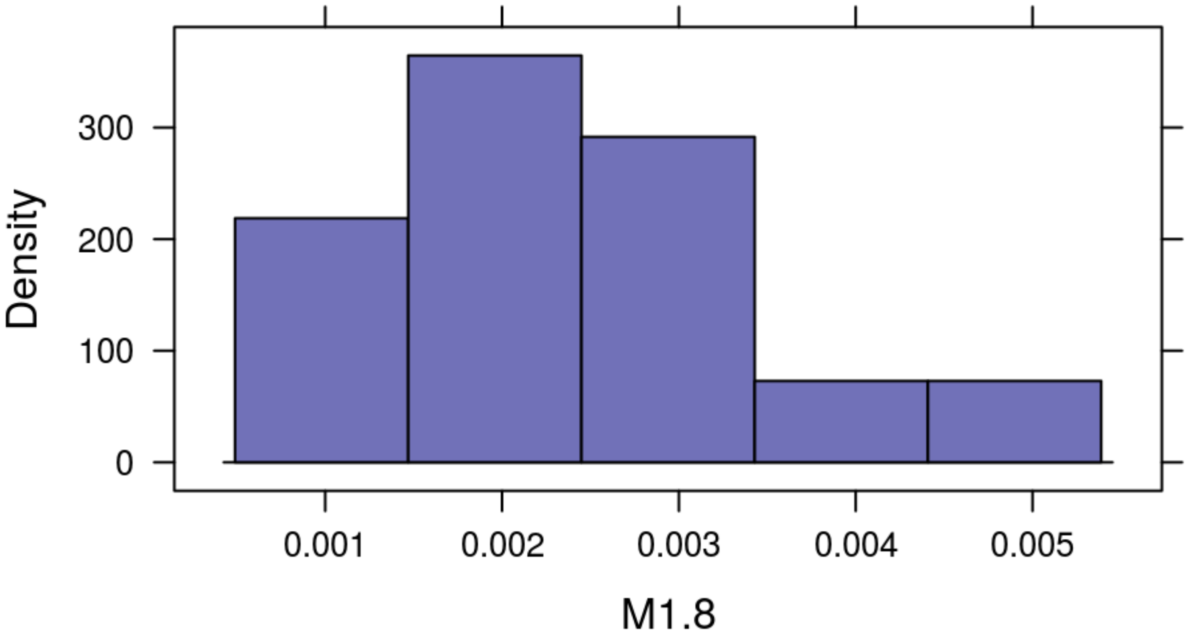

```
histogram(~M1.5,data=GenderResearch)
```



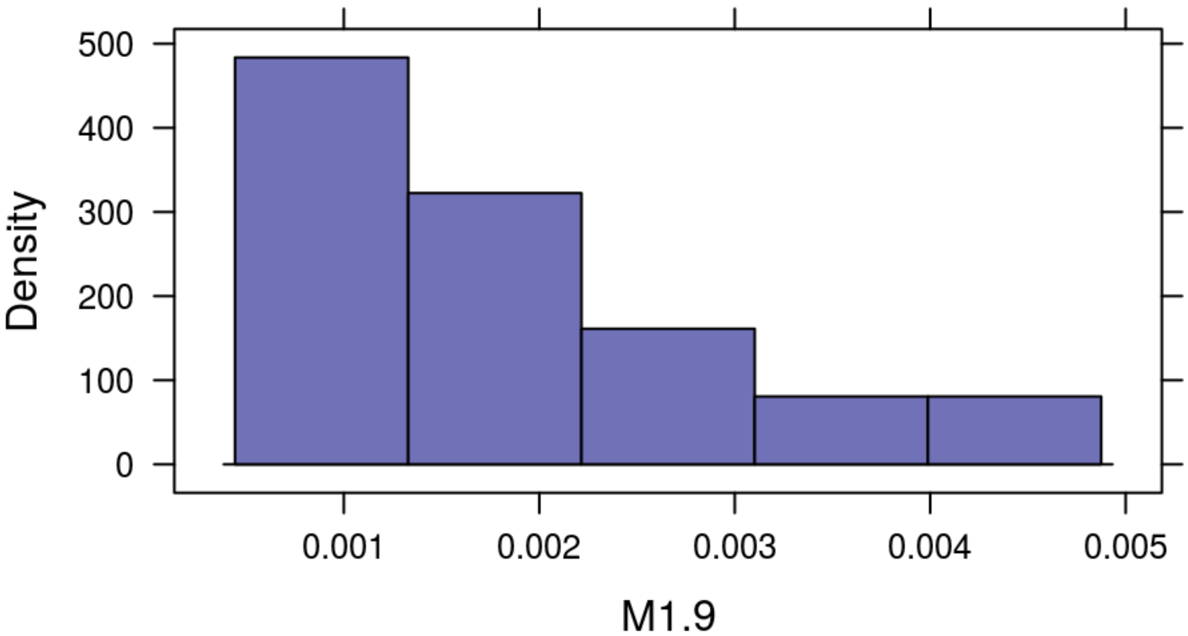

```
histogram(~M1.7,data=GenderResearch)
```



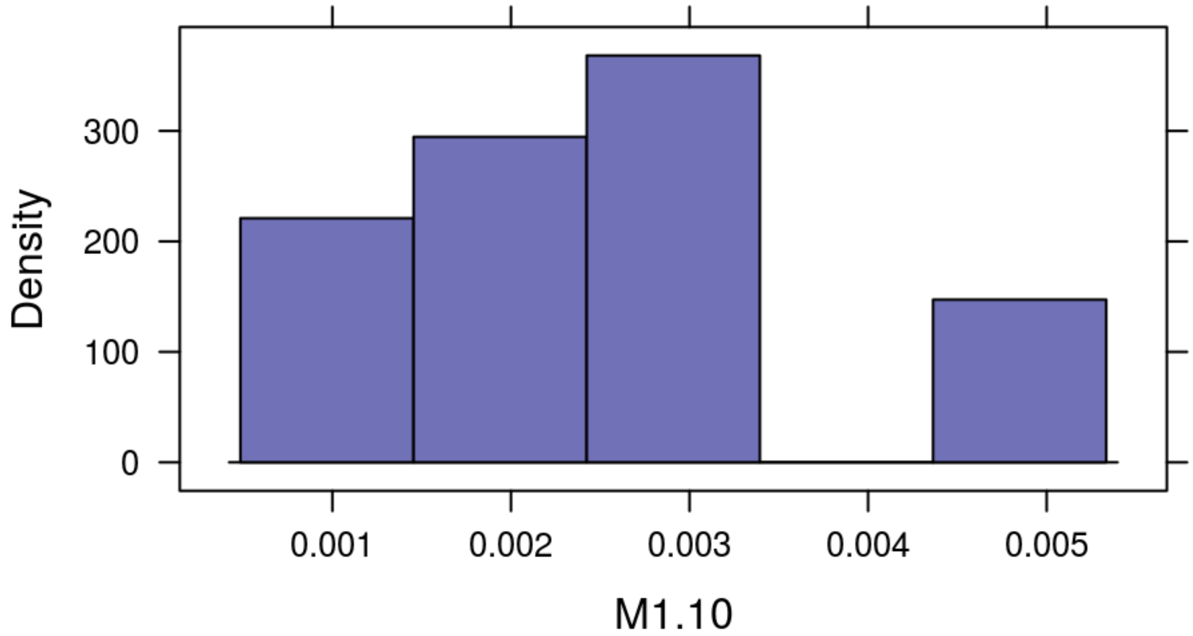

```
histogram(~M1.8,data=GenderResearch)
```



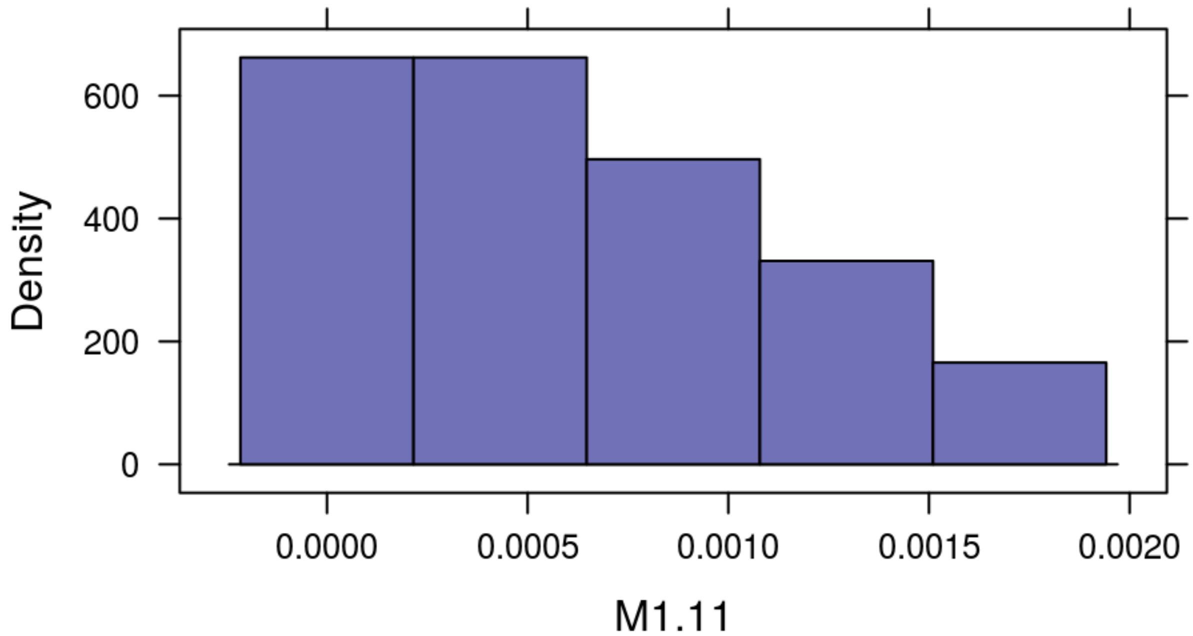

```
histogram(~M1.9,data=GenderResearch)
```



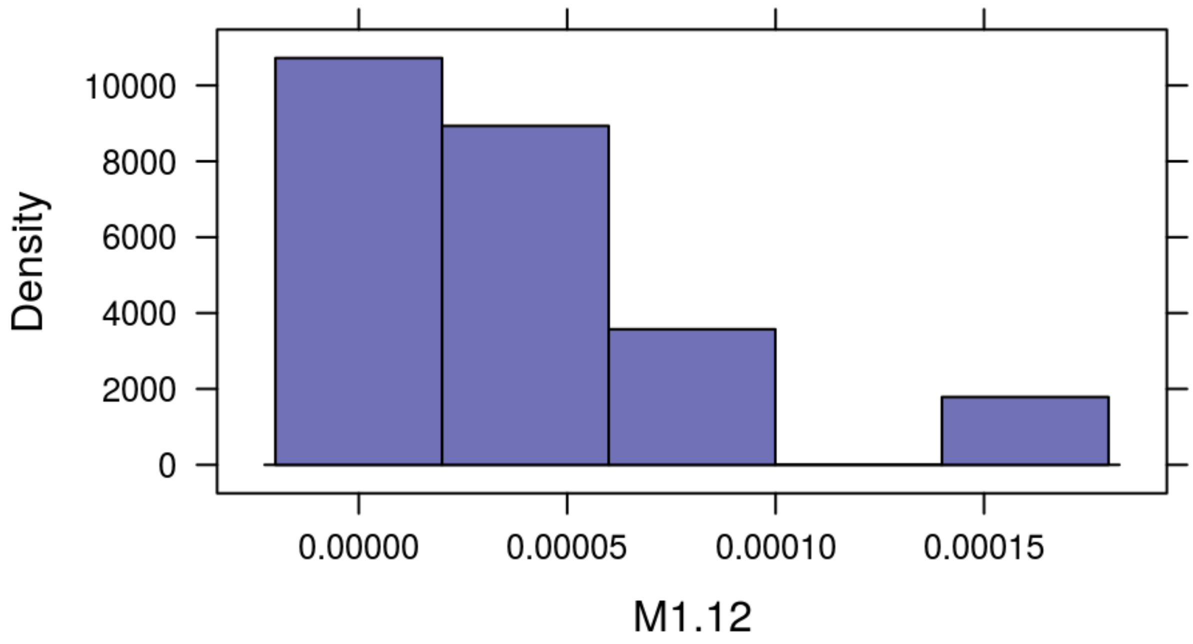

```
histogram(~M1.10,data=GenderResearch)
```



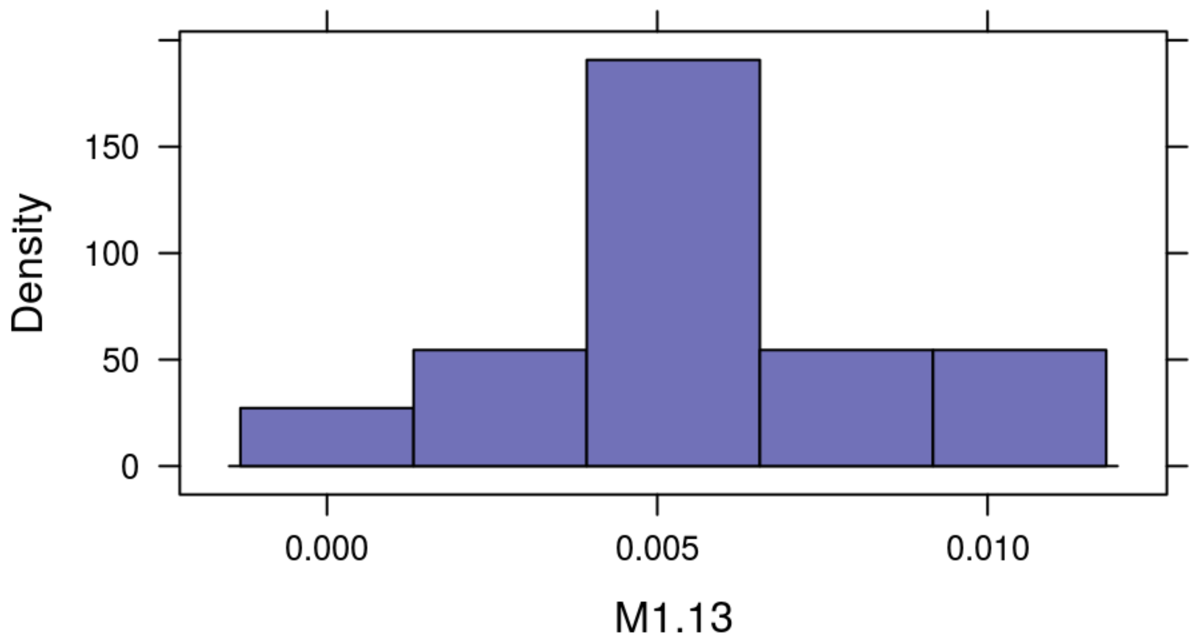

```
histogram(~M1.11,data=GenderResearch)
```



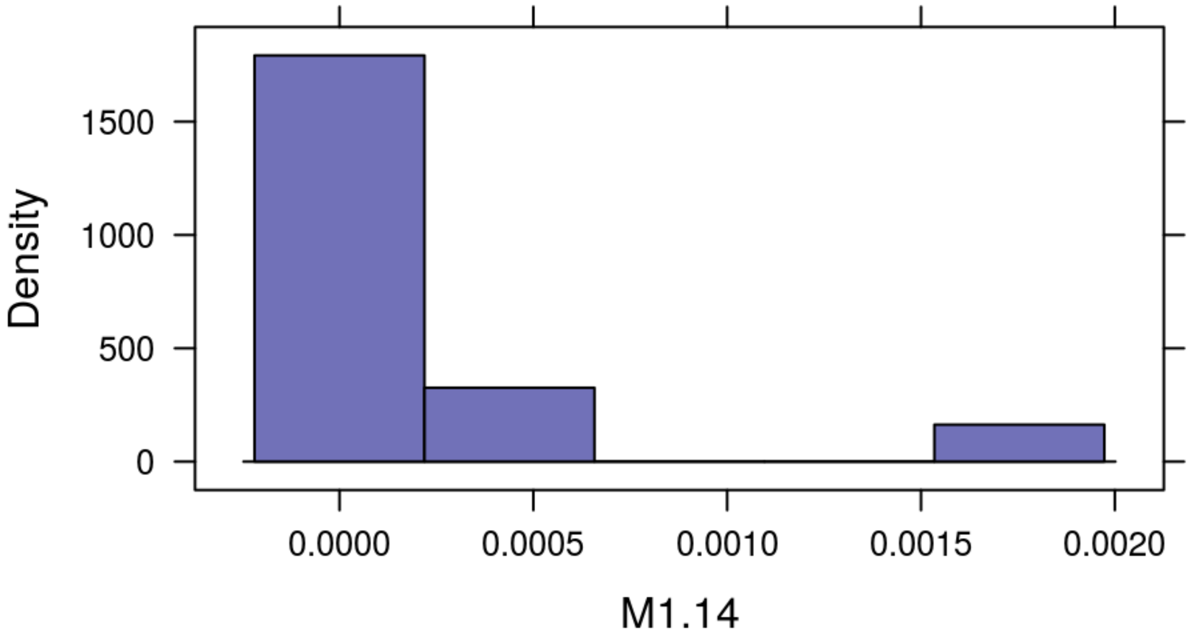

```
histogram(~M1.12,data=GenderResearch)
```



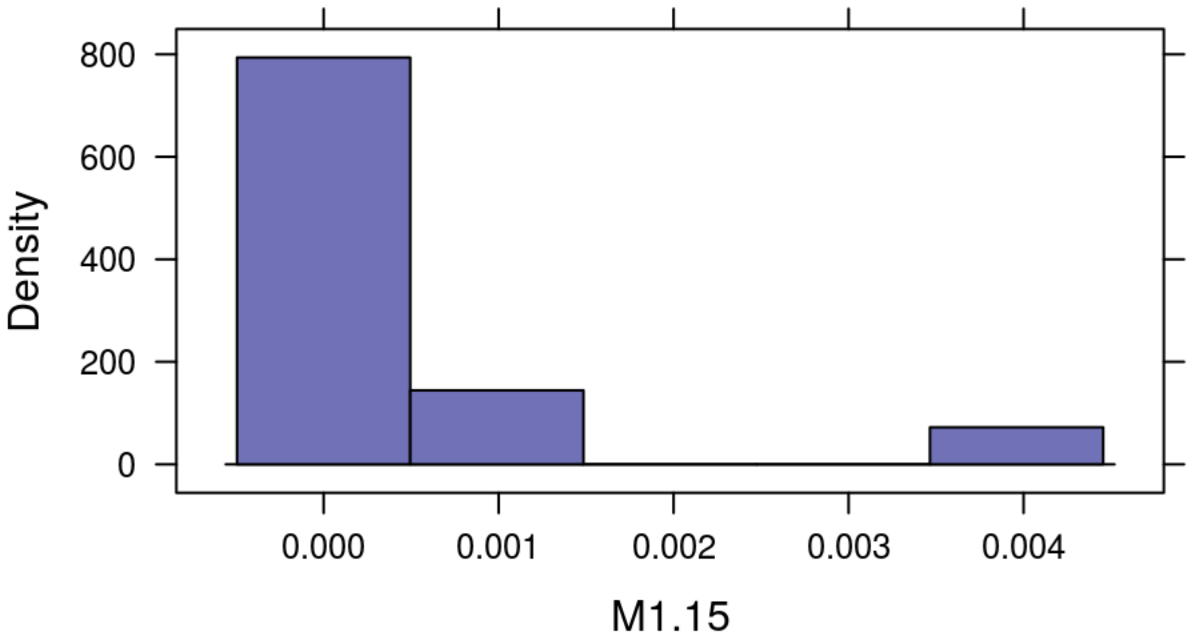

```
histogram(~M1.13,data=GenderResearch)
```



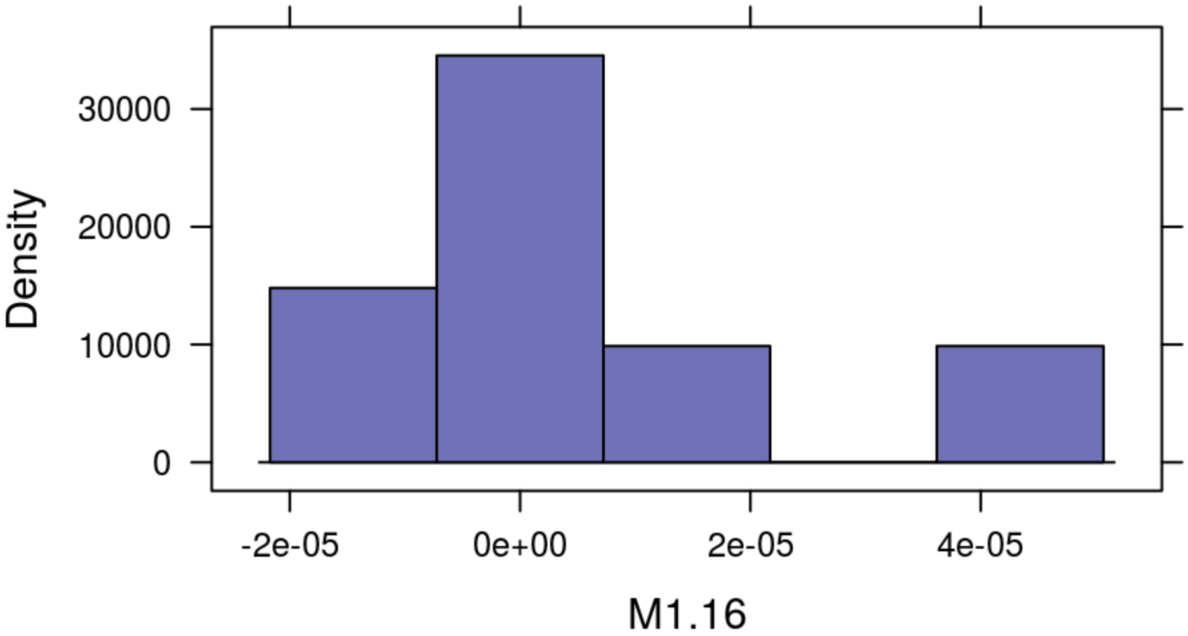

```
histogram(~M1.14,data=GenderResearch)
```



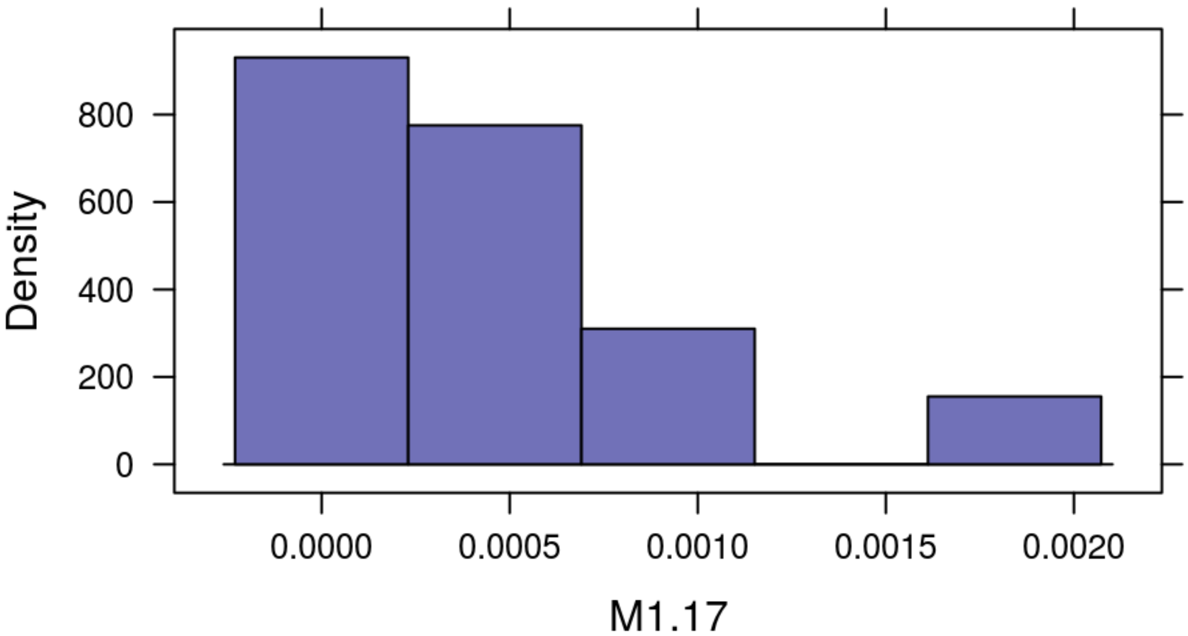

```
histogram(~M1.15,data=GenderResearch)
```



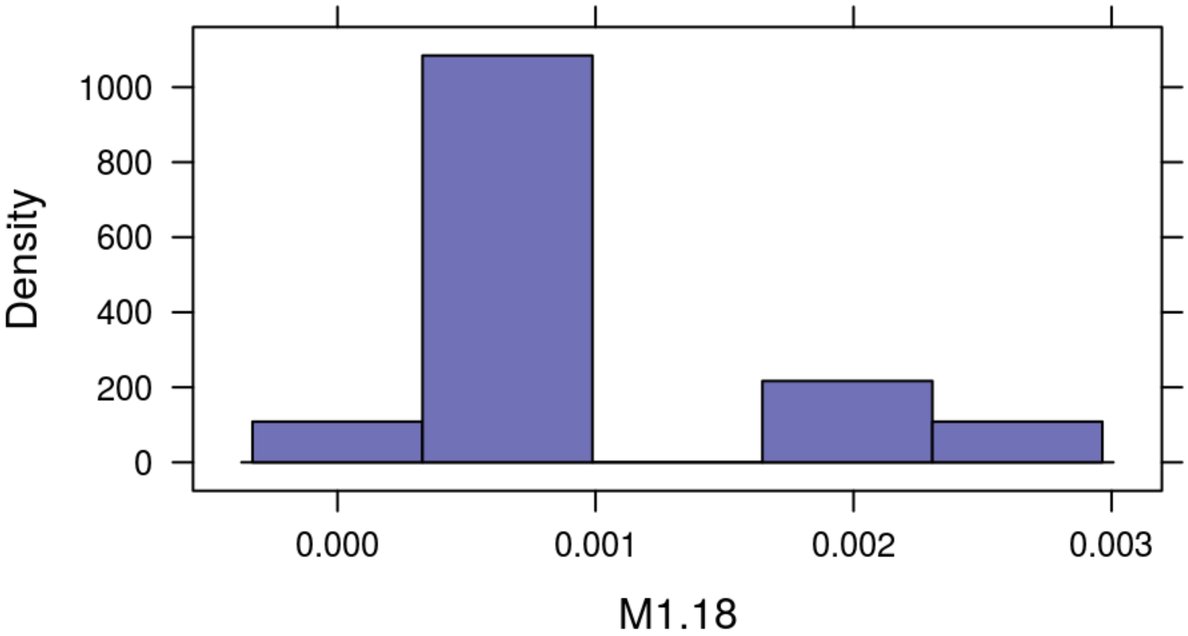

```
histogram(~M1.16,data=GenderResearch)
```



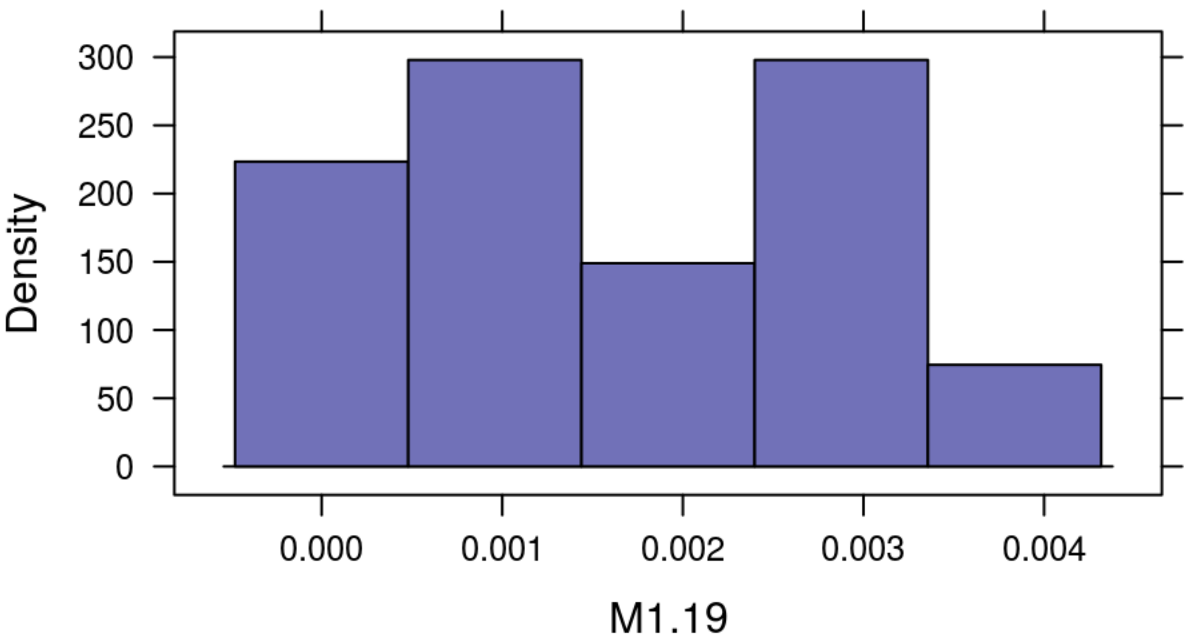

```
histogram(~M1.17,data=GenderResearch)
```



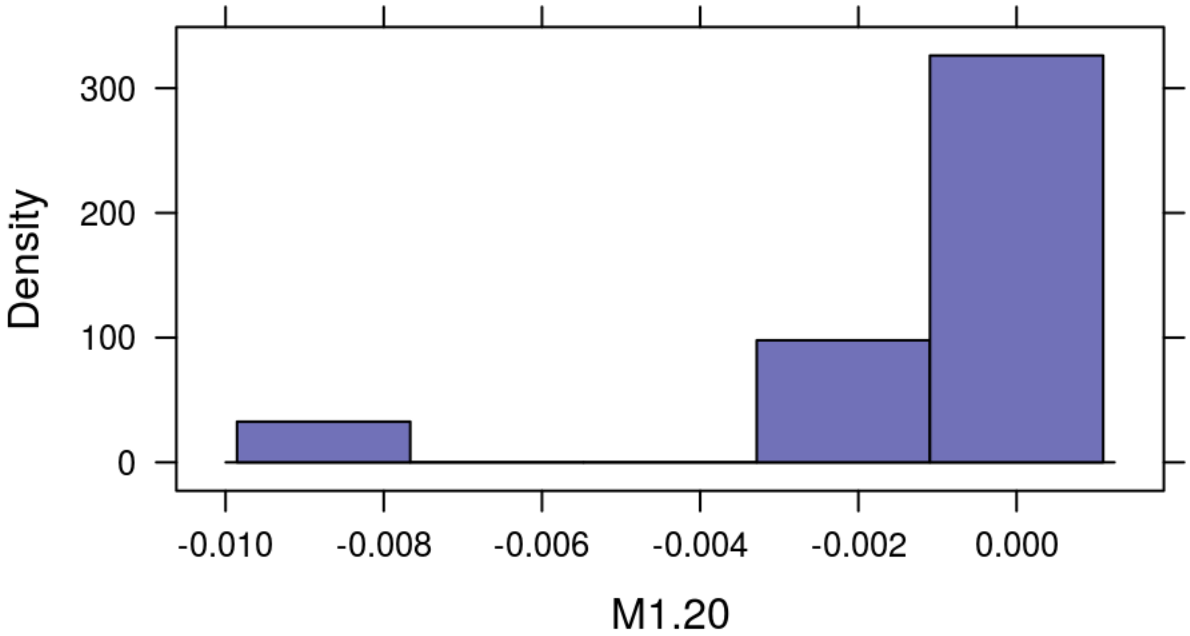

```
histogram(~M1.18,data=GenderResearch)
```



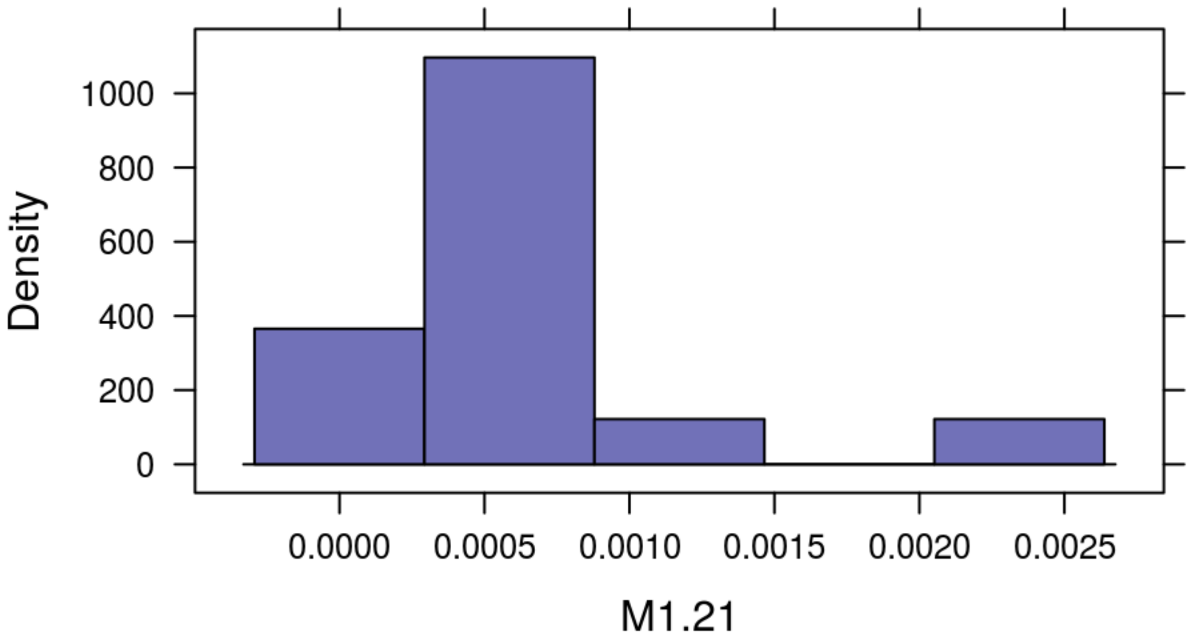

```
histogram(~M1.19,data=GenderResearch)
```



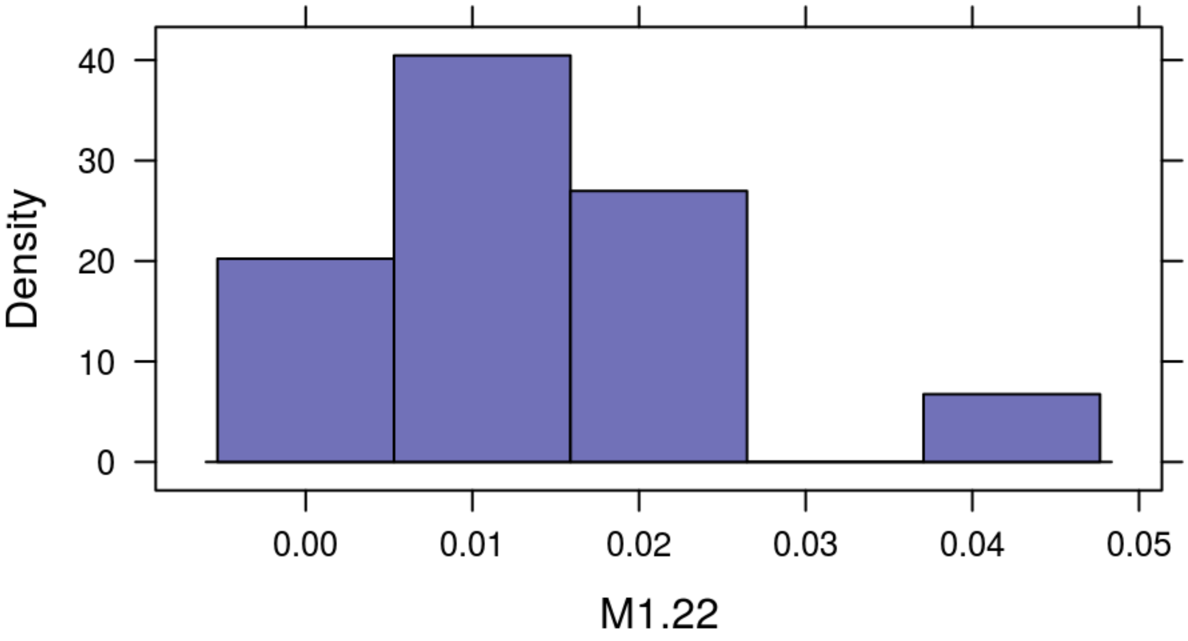

```
histogram(~M1.20,data=GenderResearch)
```



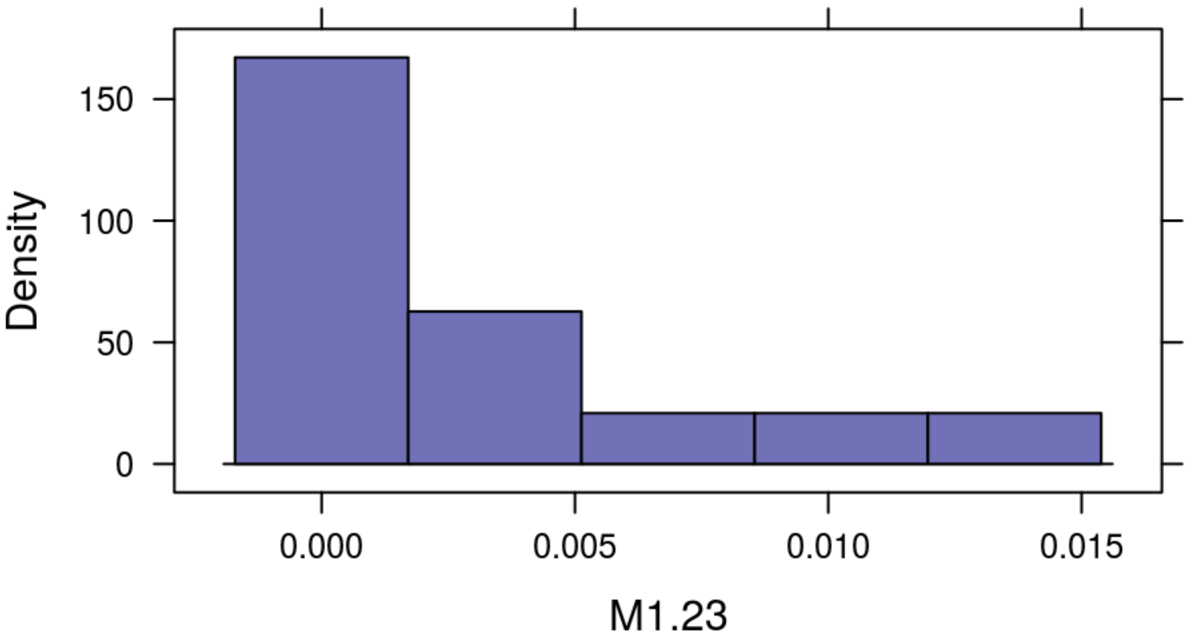

```
histogram(~M1.21,data=GenderResearch)
```



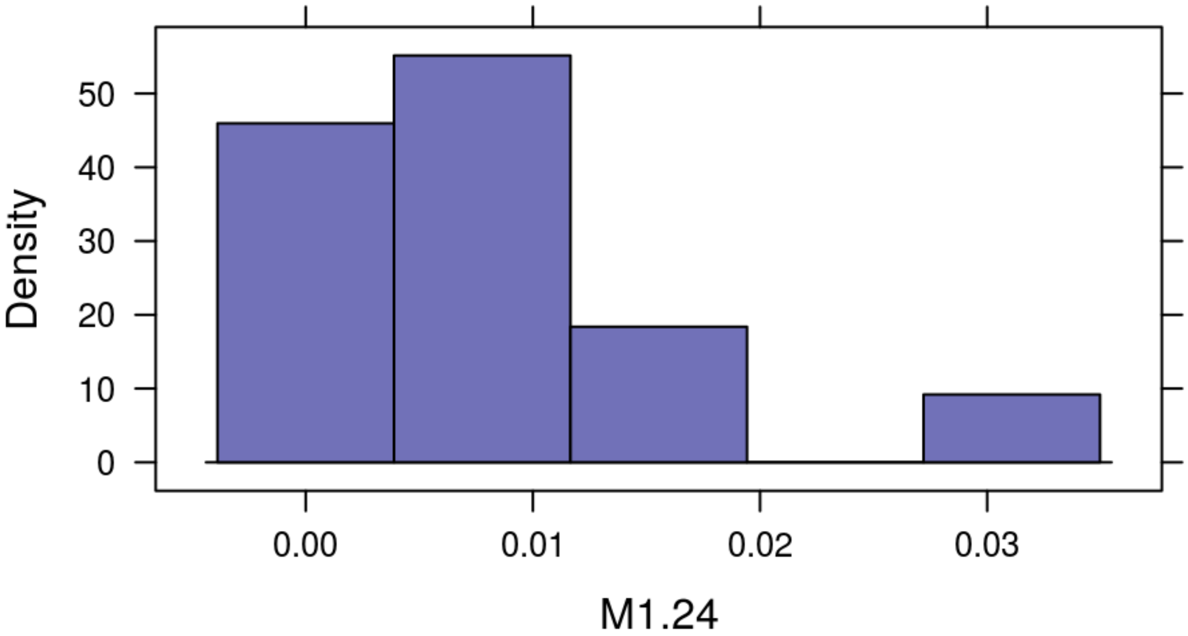

```
histogram(~M1.22,data=GenderResearch)
```



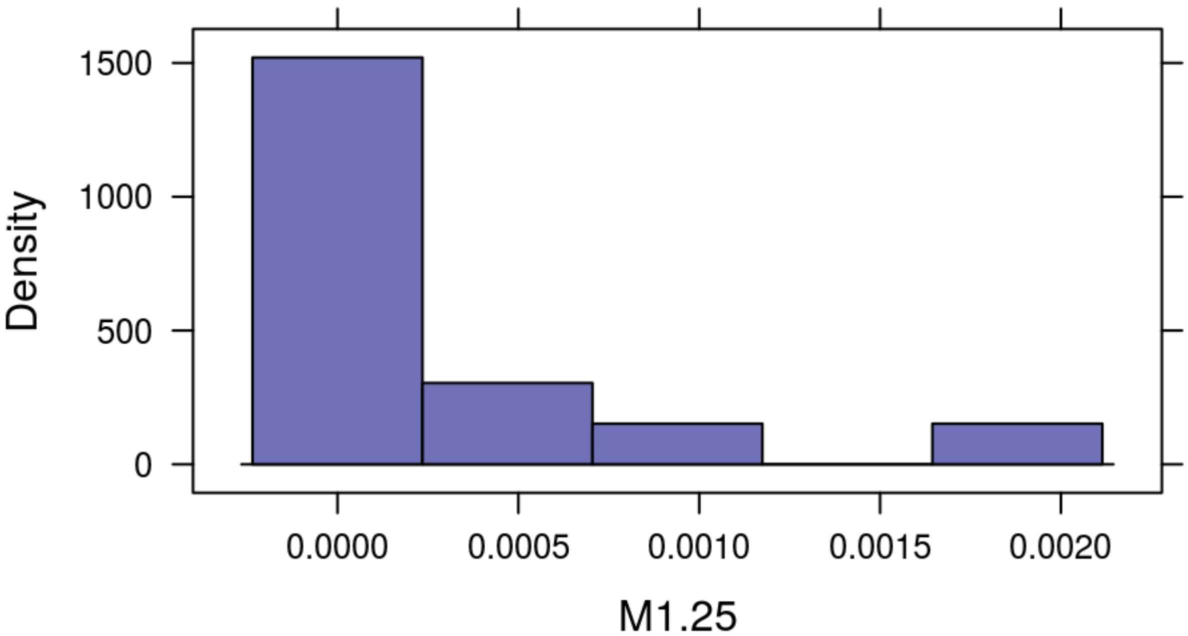

```
histogram(~M1.23,data=GenderResearch)
```



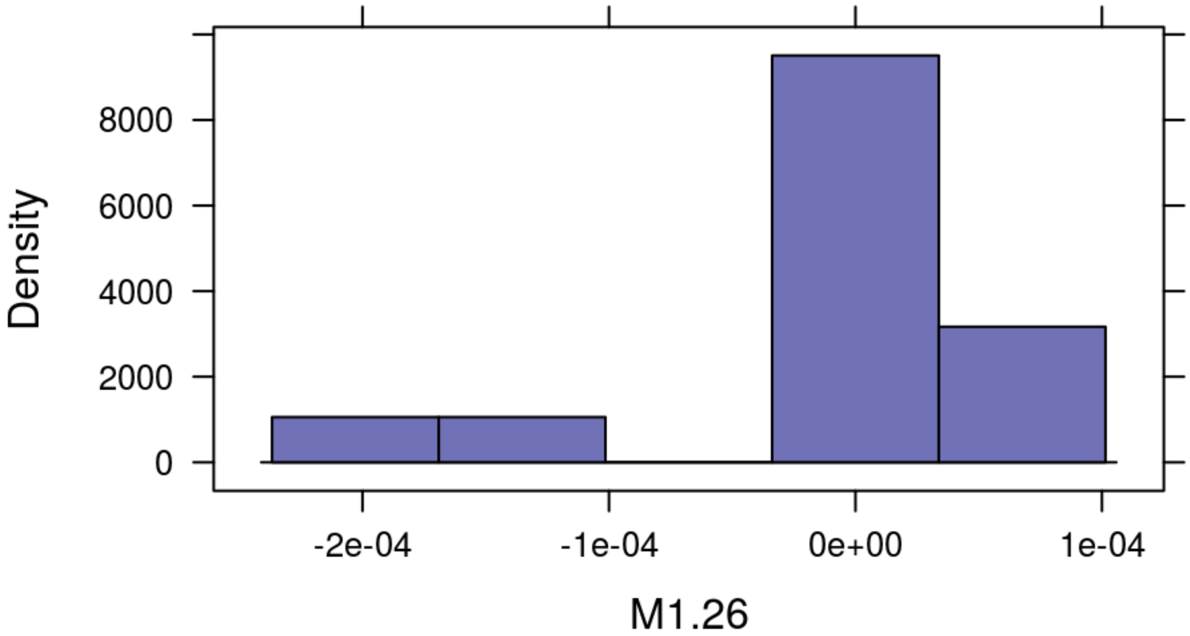

```
histogram(~M1.24,data=GenderResearch)
```



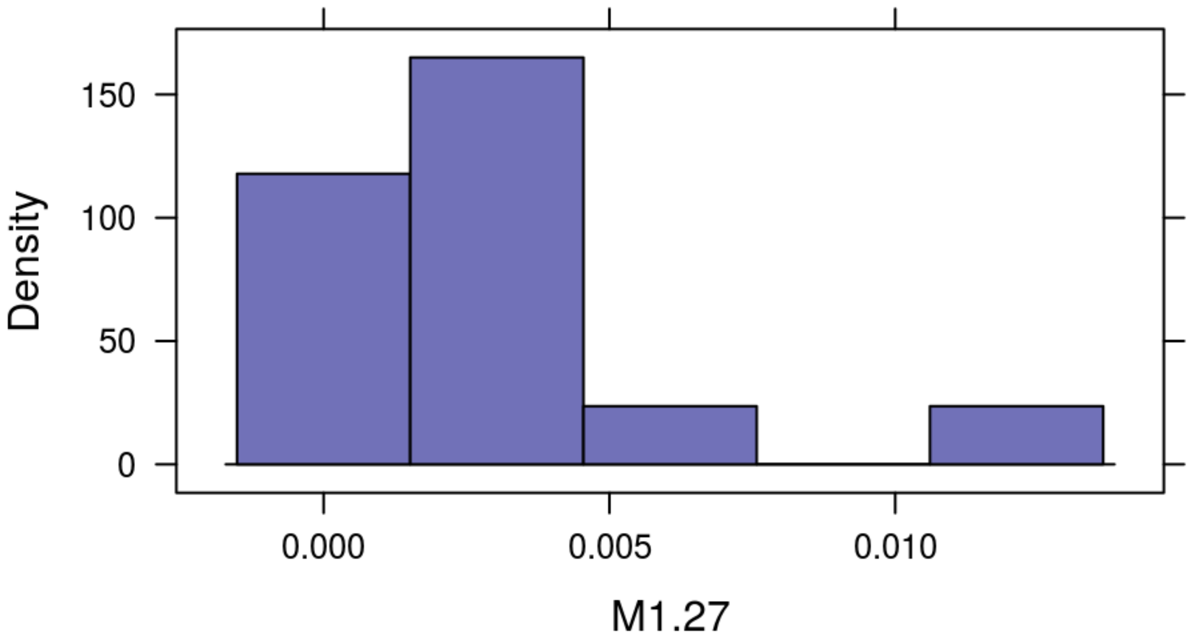

```
histogram(~M1.25,data=GenderResearch)
```



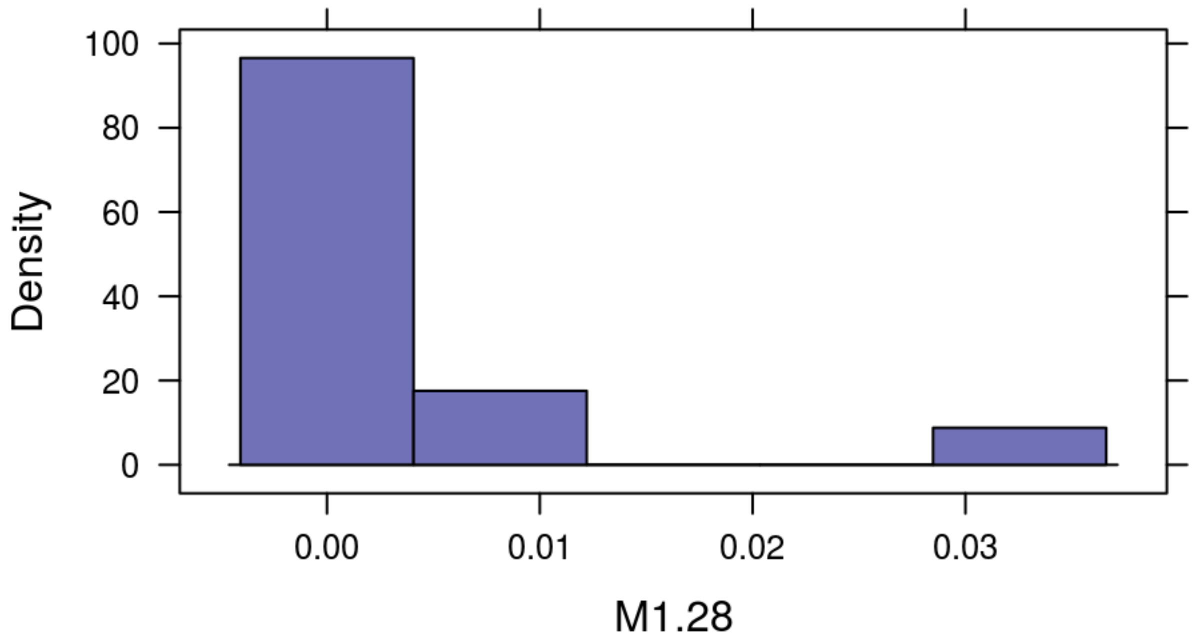

```
histogram(~M1.26,data=GenderResearch)
```



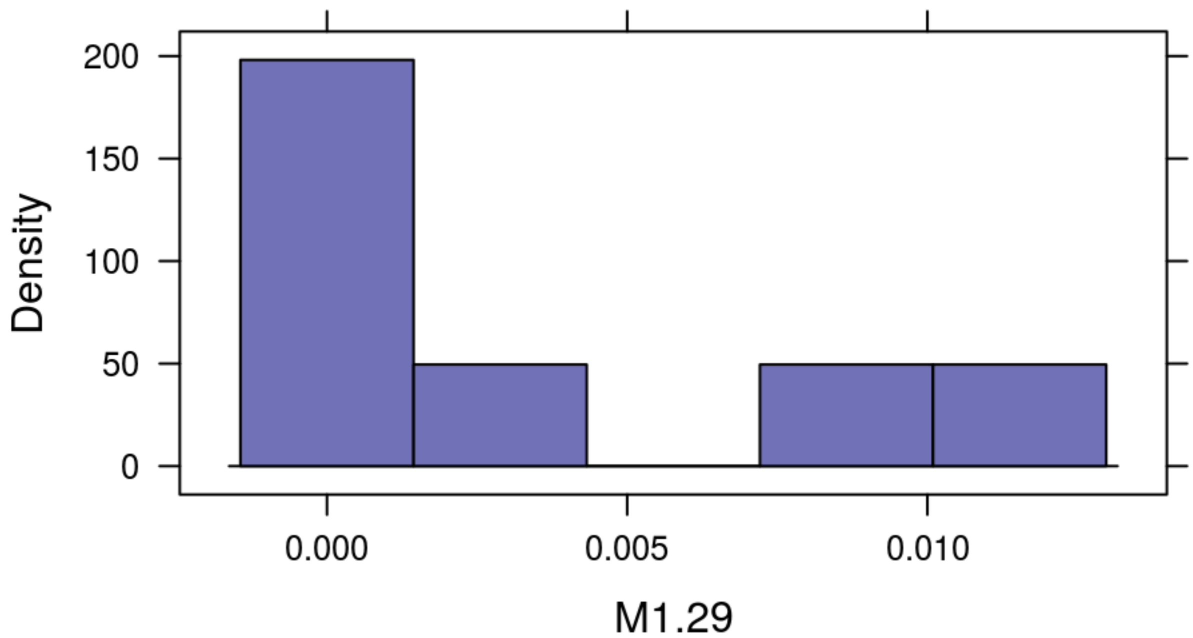

```
histogram(~M1.27,data=GenderResearch)
```



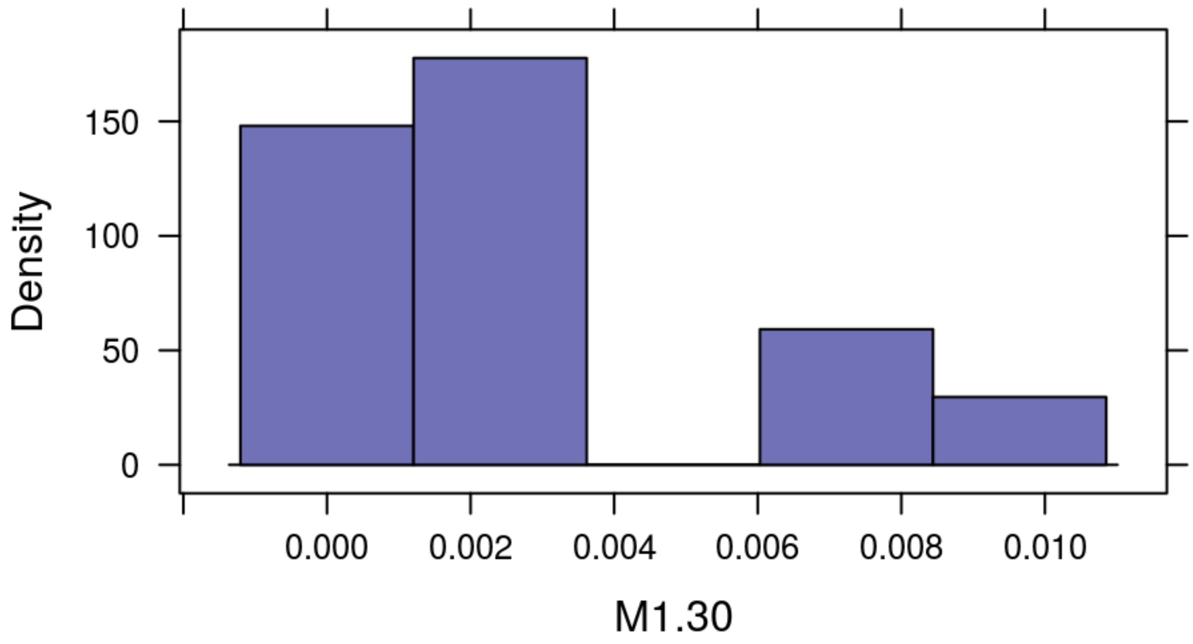

```
histogram(~M1.28,data=GenderResearch)
```



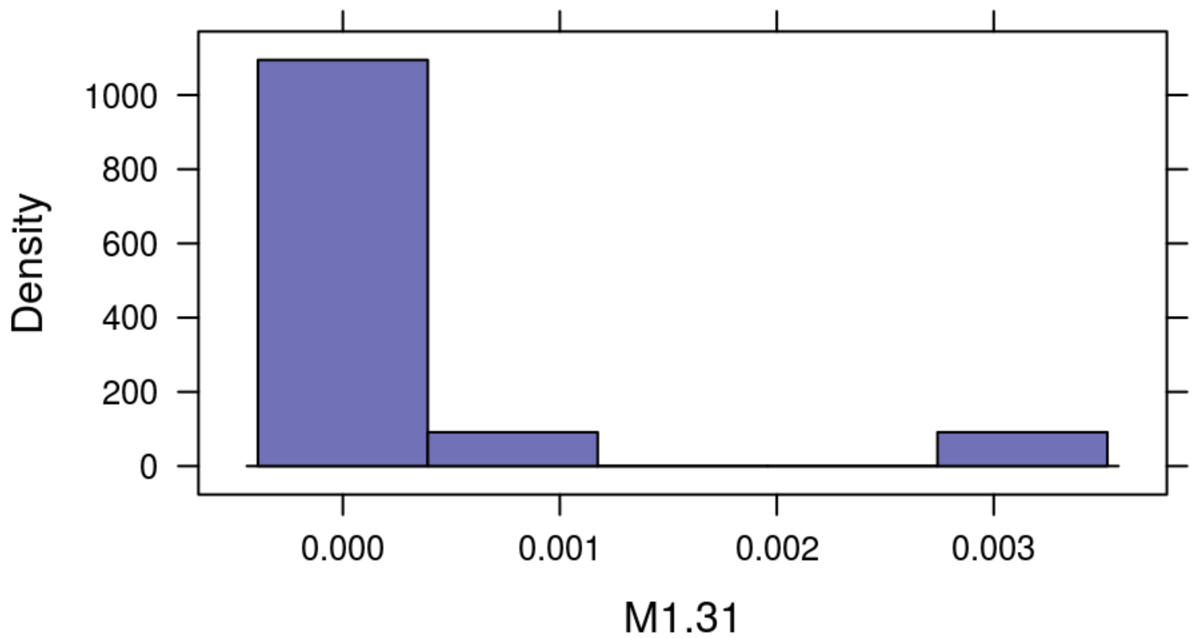

```
histogram(~M1.29,data=GenderResearch)
```



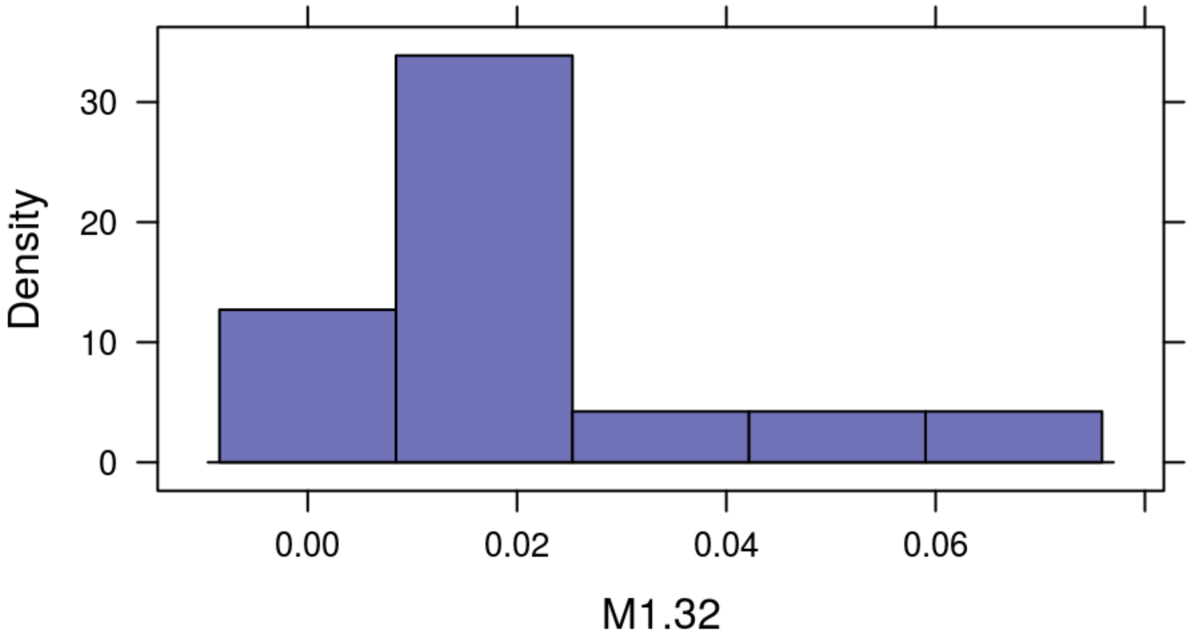

```
histogram(~M1.30,data=GenderResearch)
```



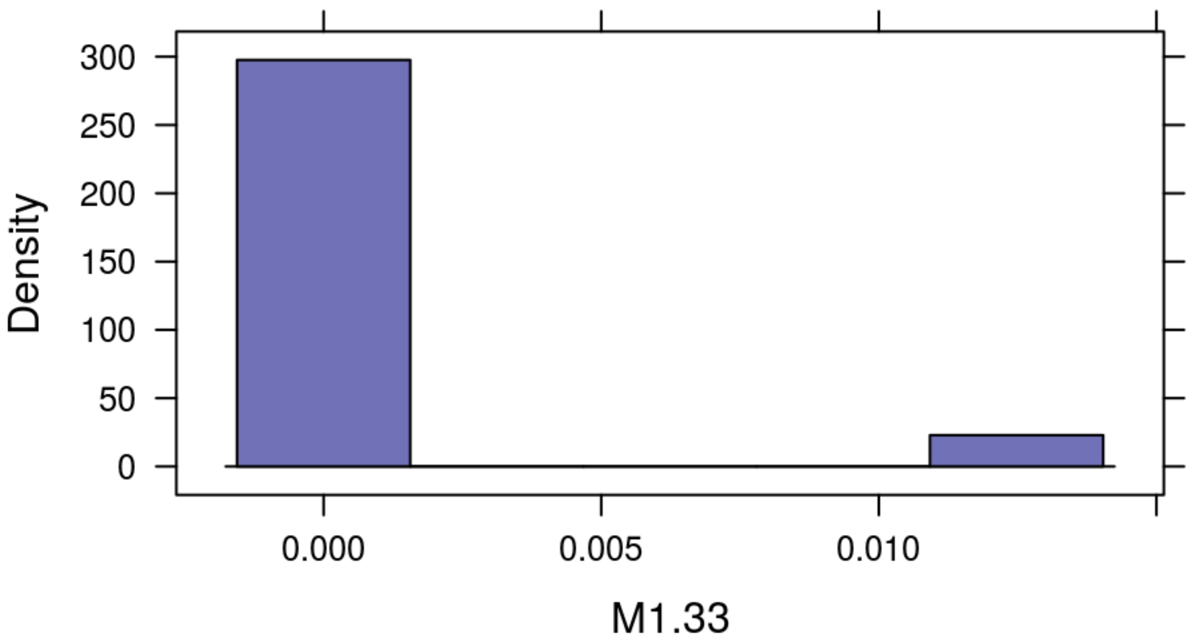

```
histogram(~M1.31,data=GenderResearch)
```



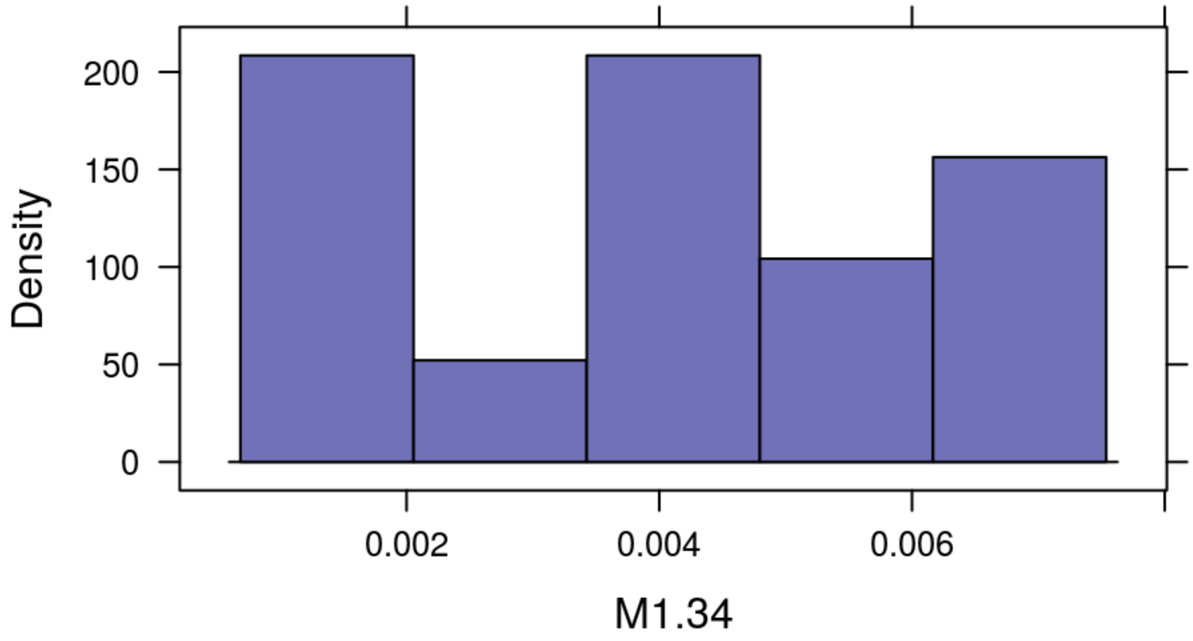

```
histogram(~M1.32,data=GenderResearch)
```



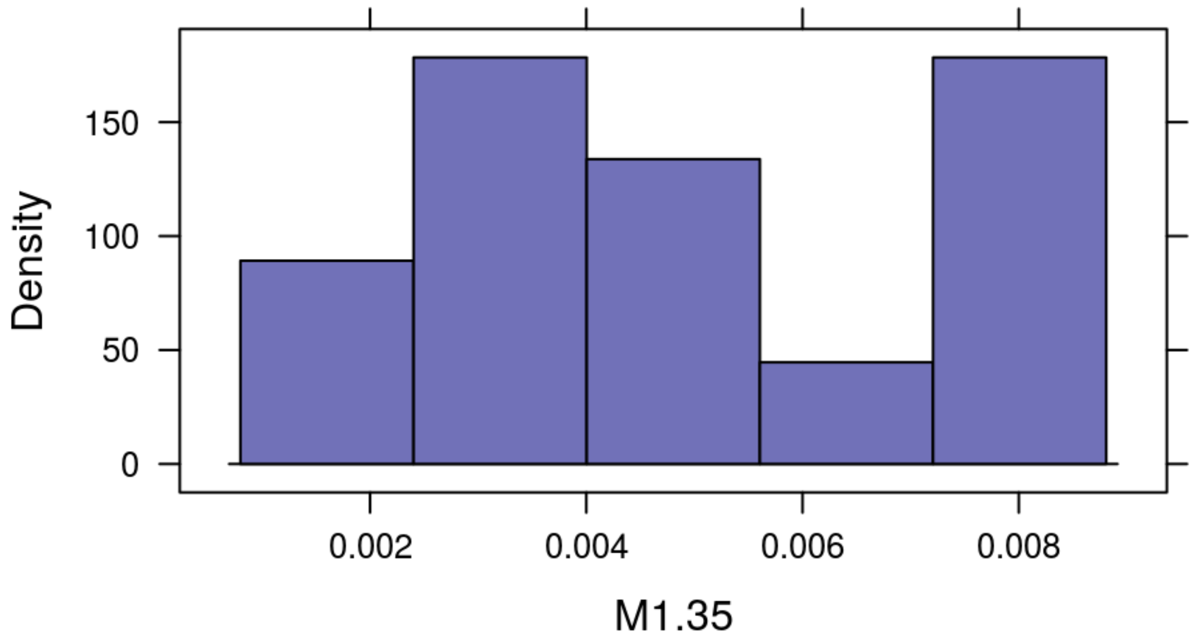

```
histogram(~M1.33,data=GenderResearch)
```




```
histogram(~M1.34,data=GenderResearch)
```




```
histogram(~M1.35,data=GenderResearch)
```

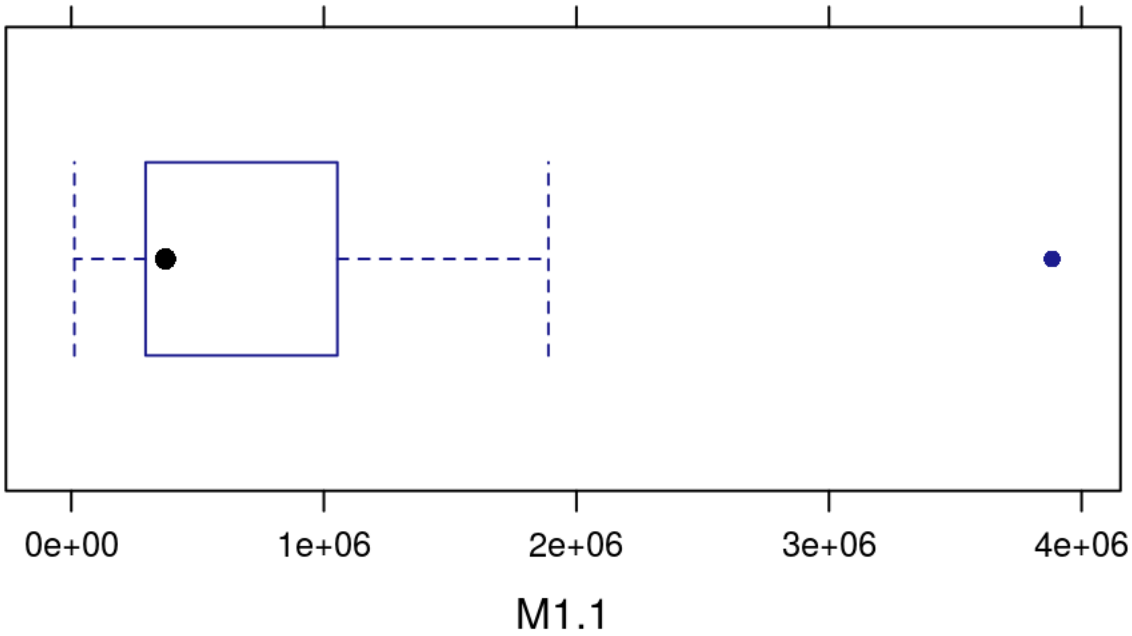


Talk about graphs here…

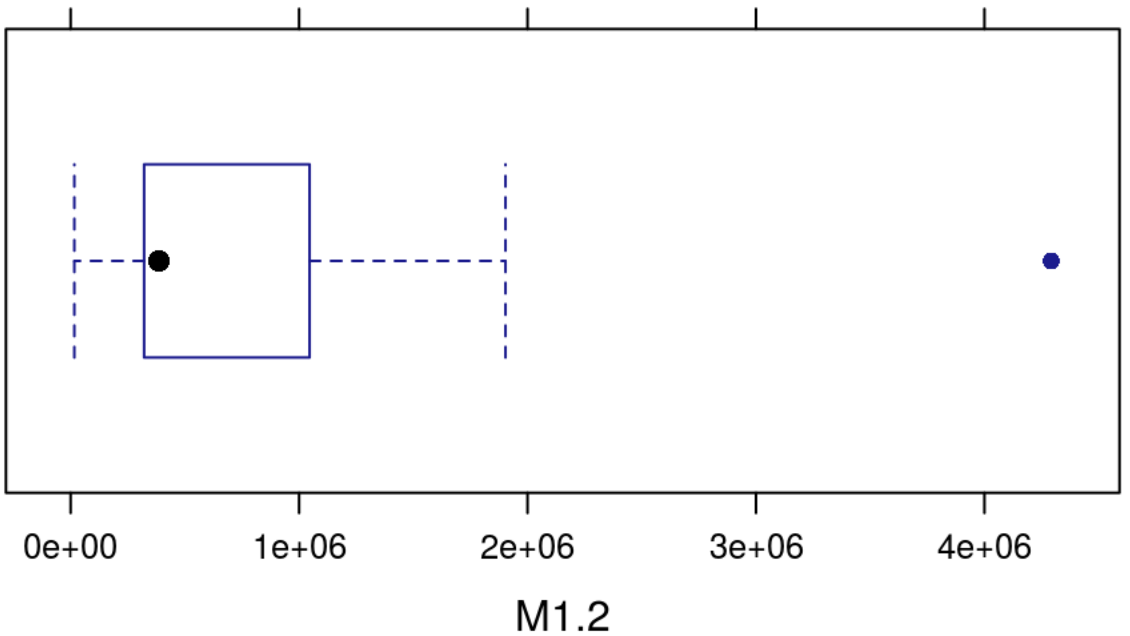
Boxplots are here…

Below are the box and whisker plots for each variable. I made these graphs to see how how big my outliers were and wihch category gave me the most outliers that way I could deicde if I wished to get rid of one of the categories to minimize the outliers. This also helped me see where the means lie even after normalizing the data.

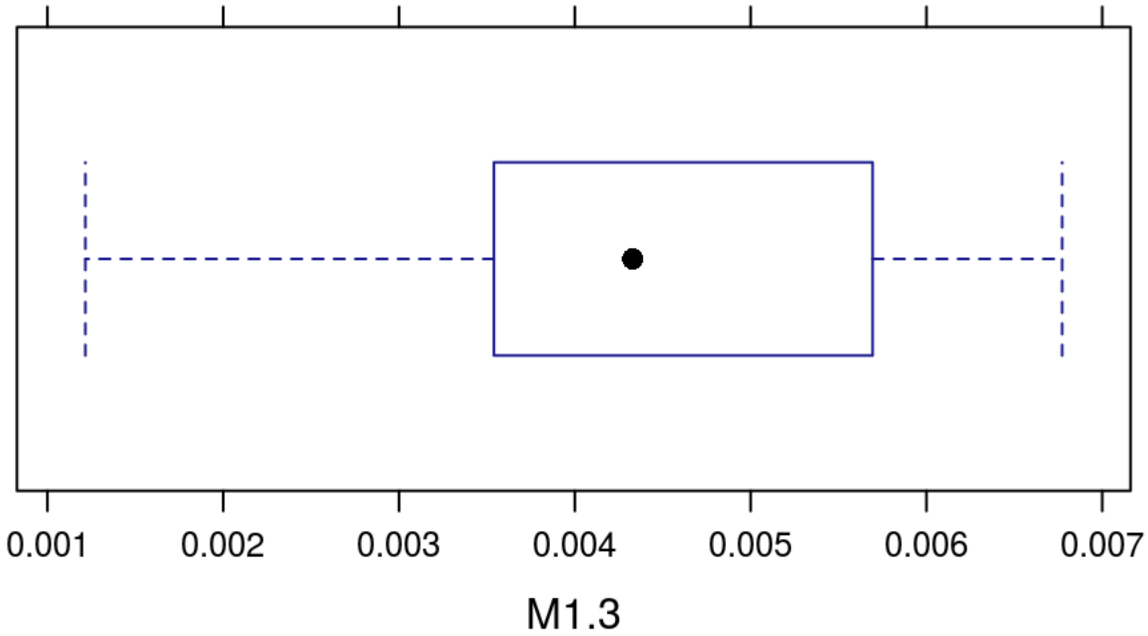
```
bwplot(~M1.1,data=GenderResearch)
```



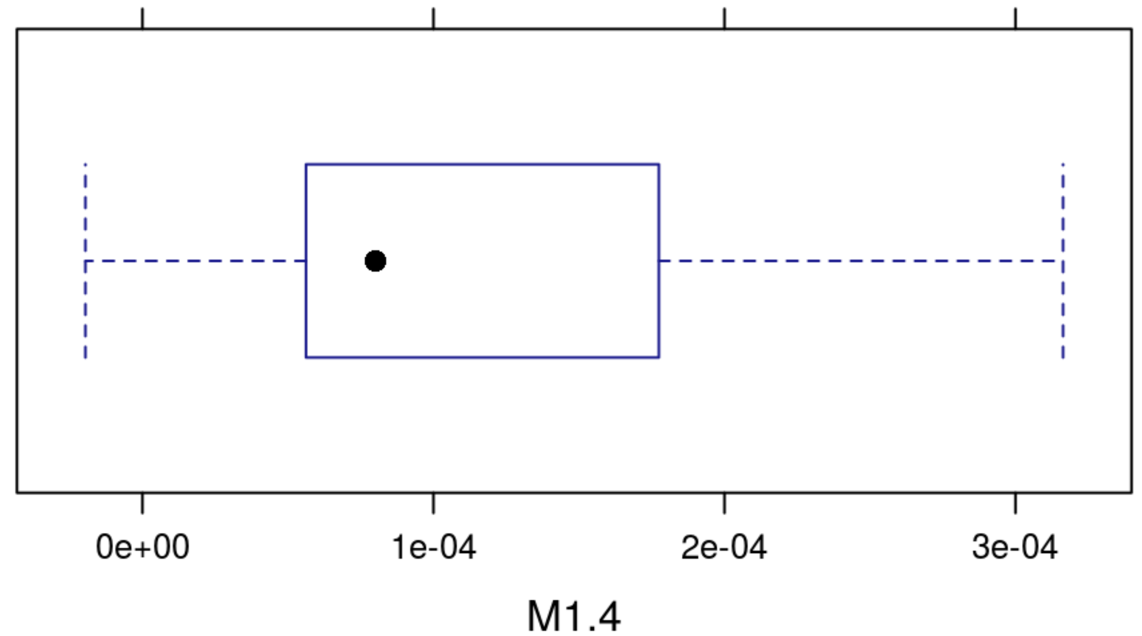

```
bwplot(~M1.2,data=GenderResearch)
```



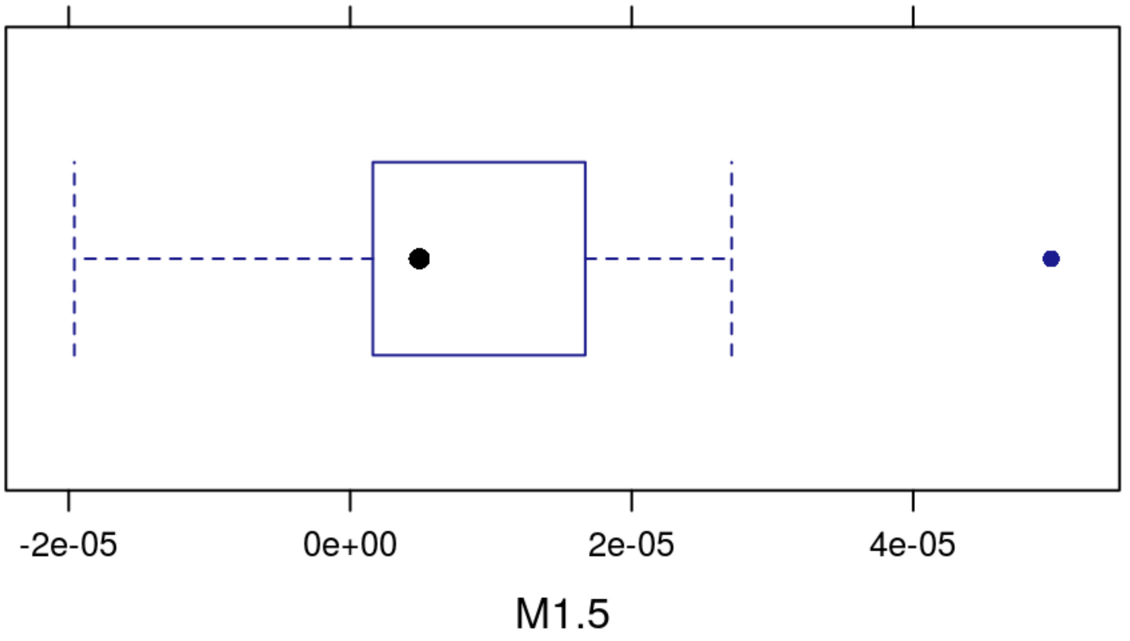

```
bwplot(~M1.3,data=GenderResearch)
```



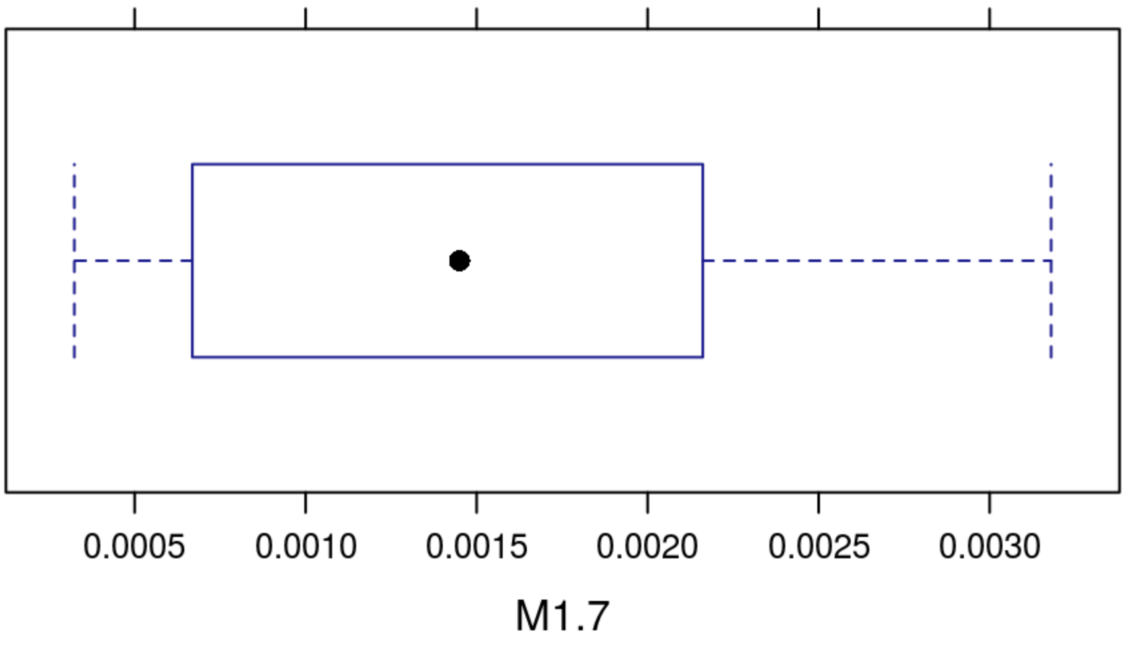

```
bwplot(~M1.4,data=GenderResearch)
```



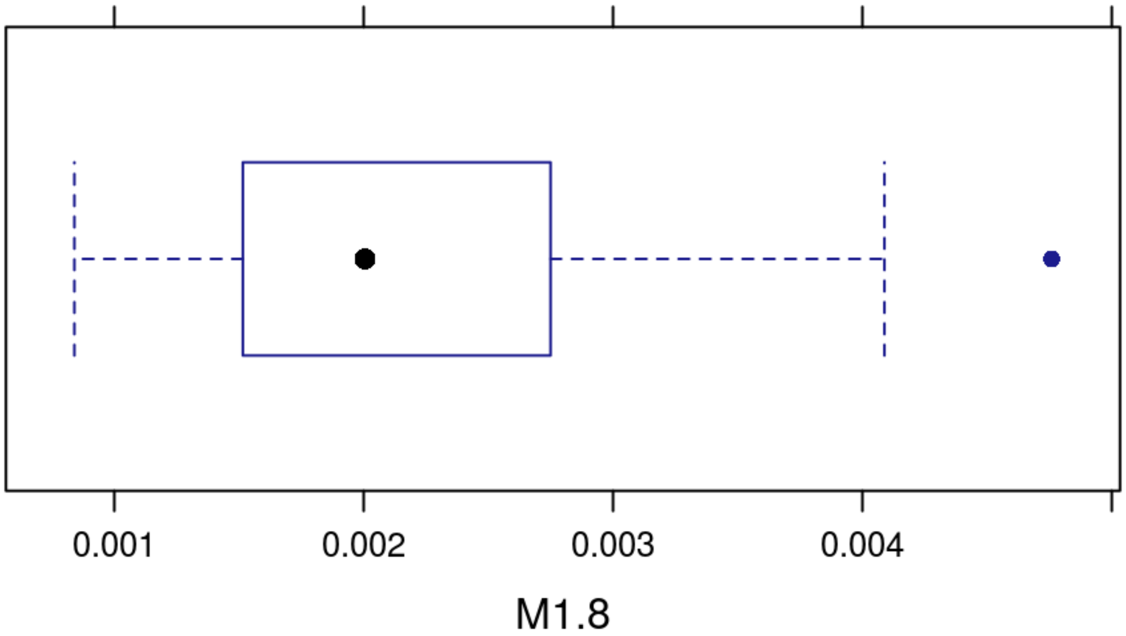

```
bwplot(~M1.5,data=GenderResearch)
```



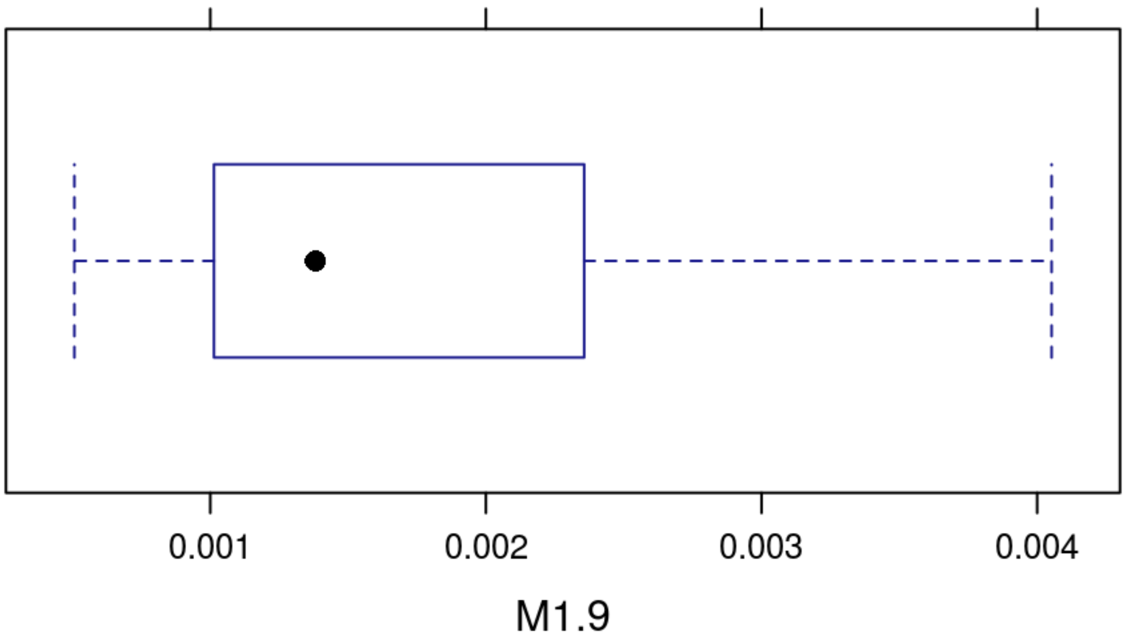

```
bwplot(~M1.7,data=GenderResearch)
```



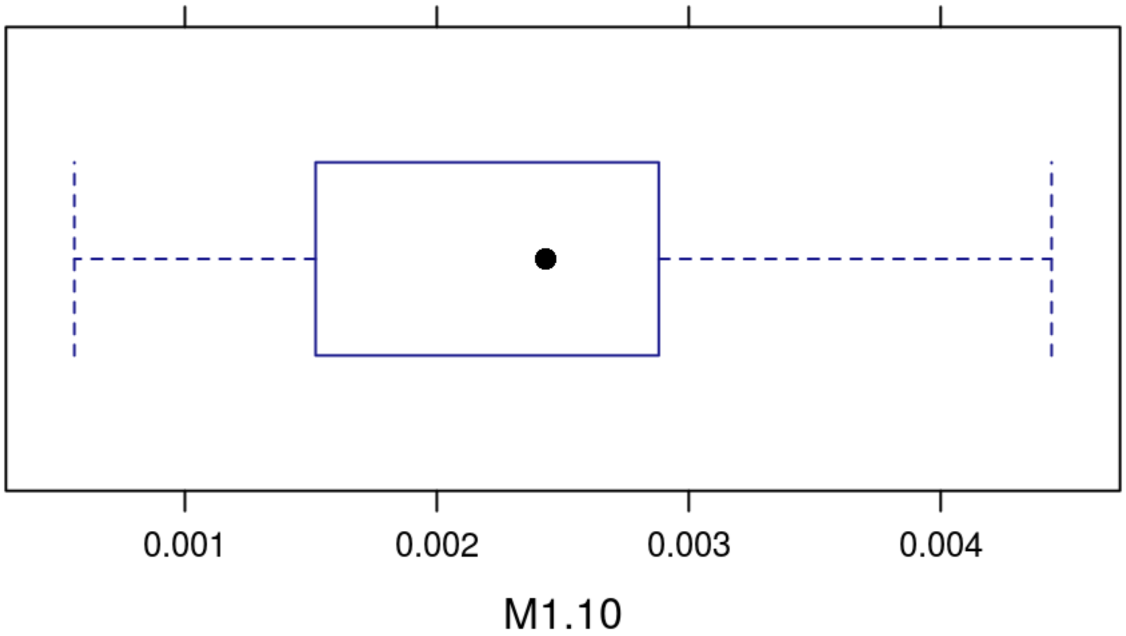

```
bwplot(~M1.8,data=GenderResearch)
```



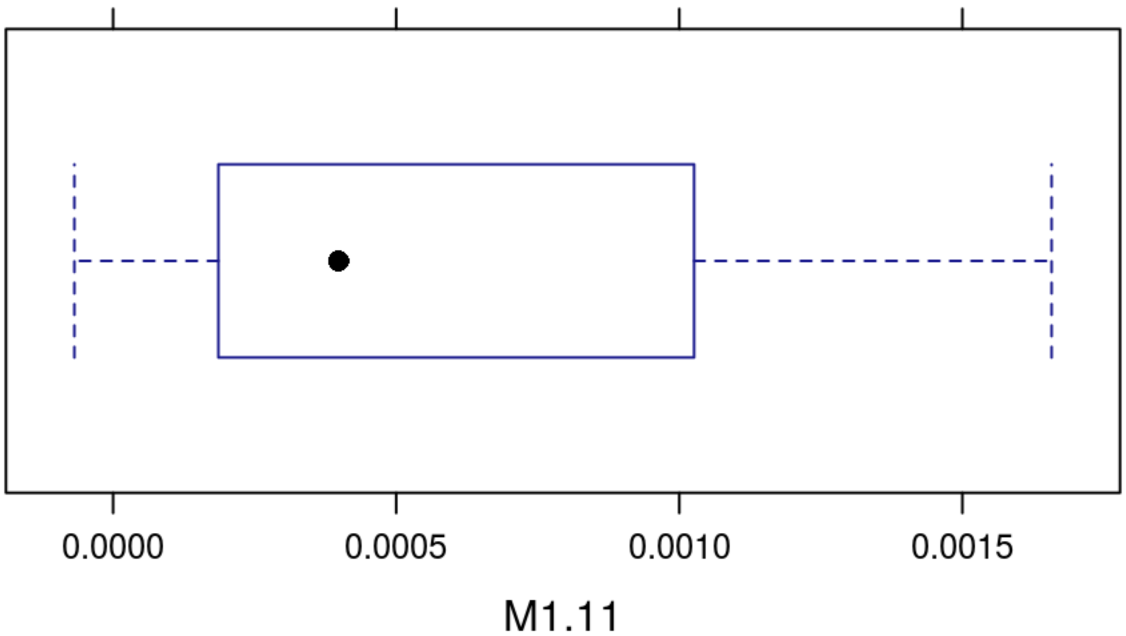

```
bwplot(~M1.9,data=GenderResearch)
```



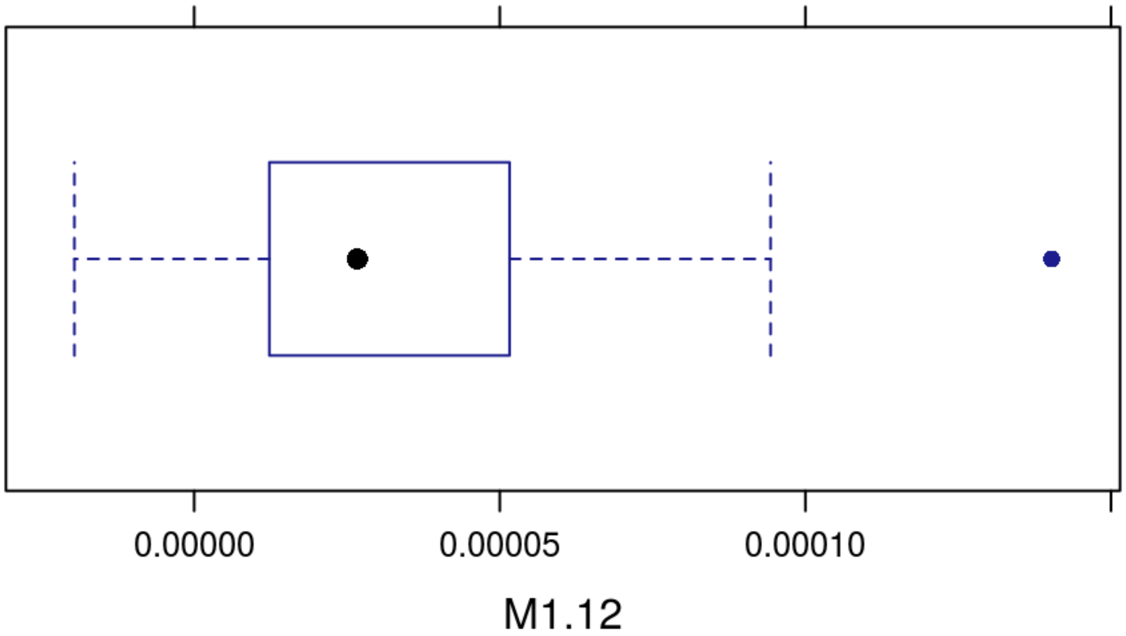

```
bwplot(~M1.10,data=GenderResearch)
```



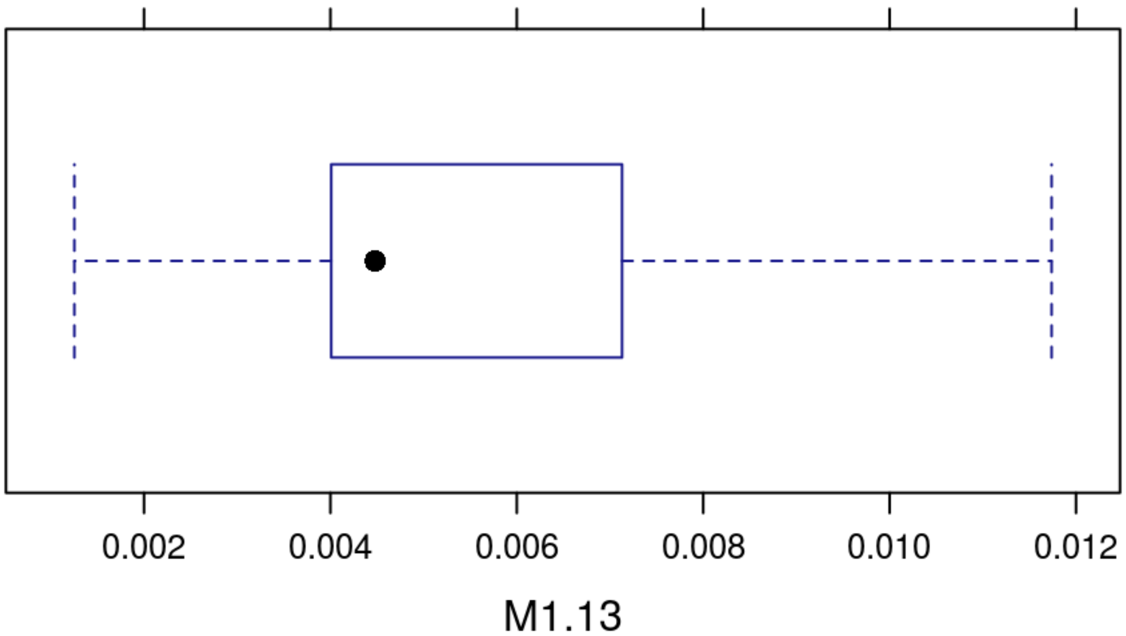

```
bwplot(~M1.11,data=GenderResearch)
```



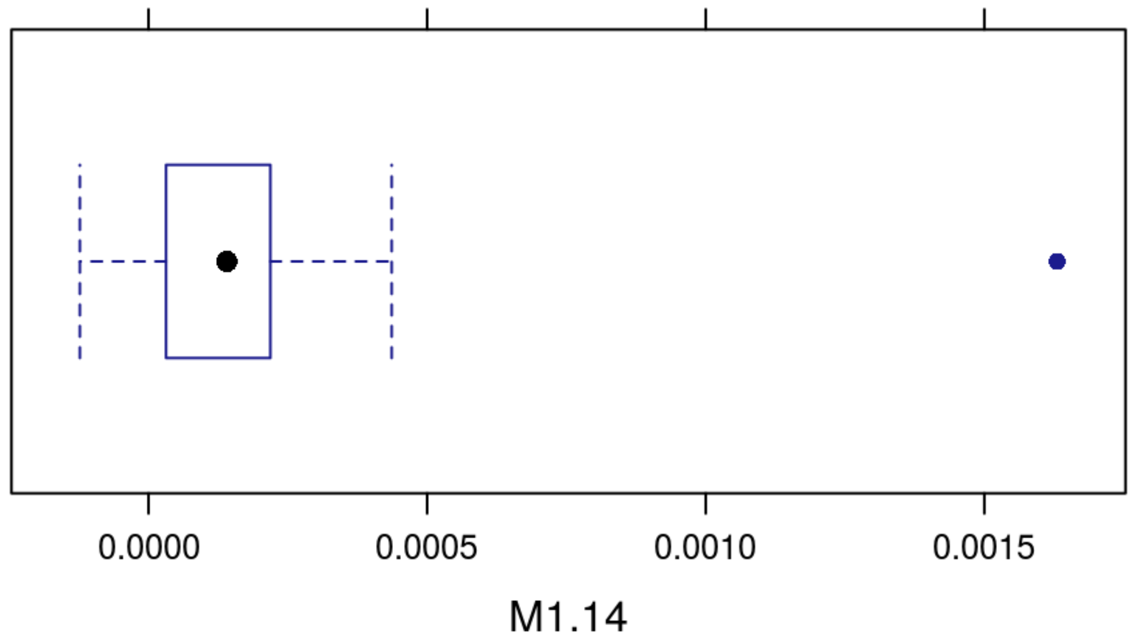

```
bwplot(~M1.12,data=GenderResearch)
```



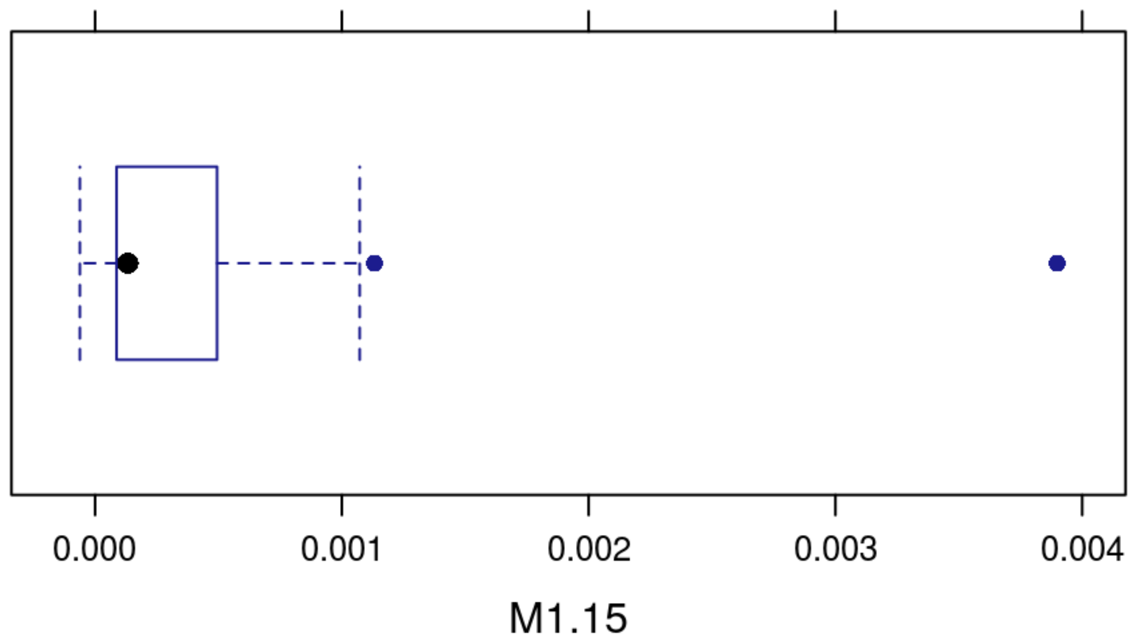

```
bwplot(~M1.13,data=GenderResearch)
```



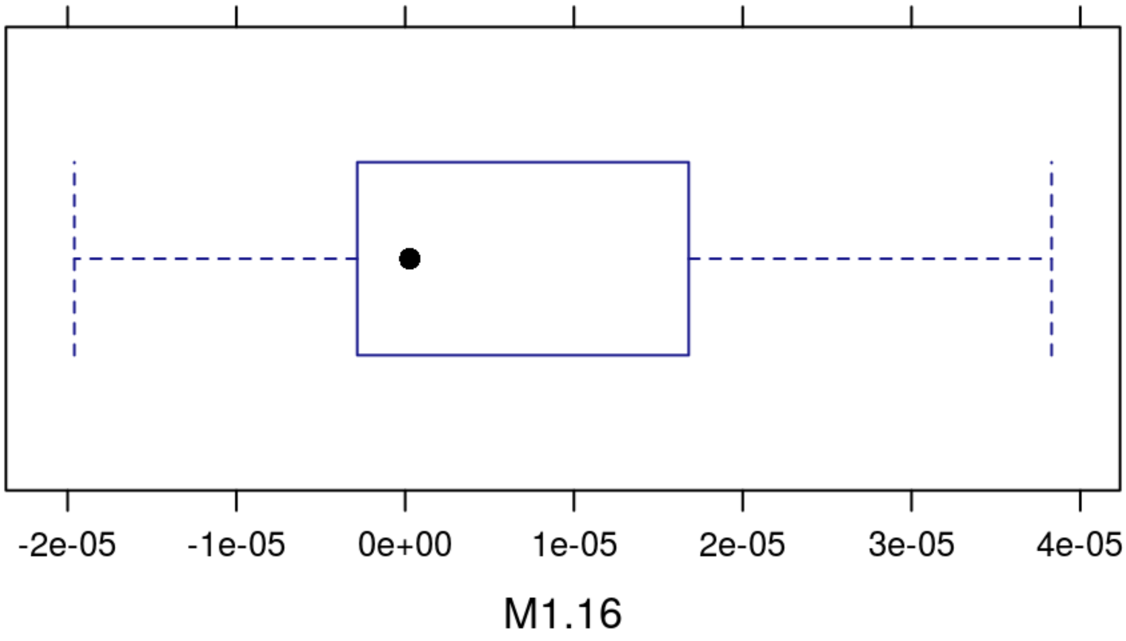

```
bwplot(~M1.14,data=GenderResearch)
```



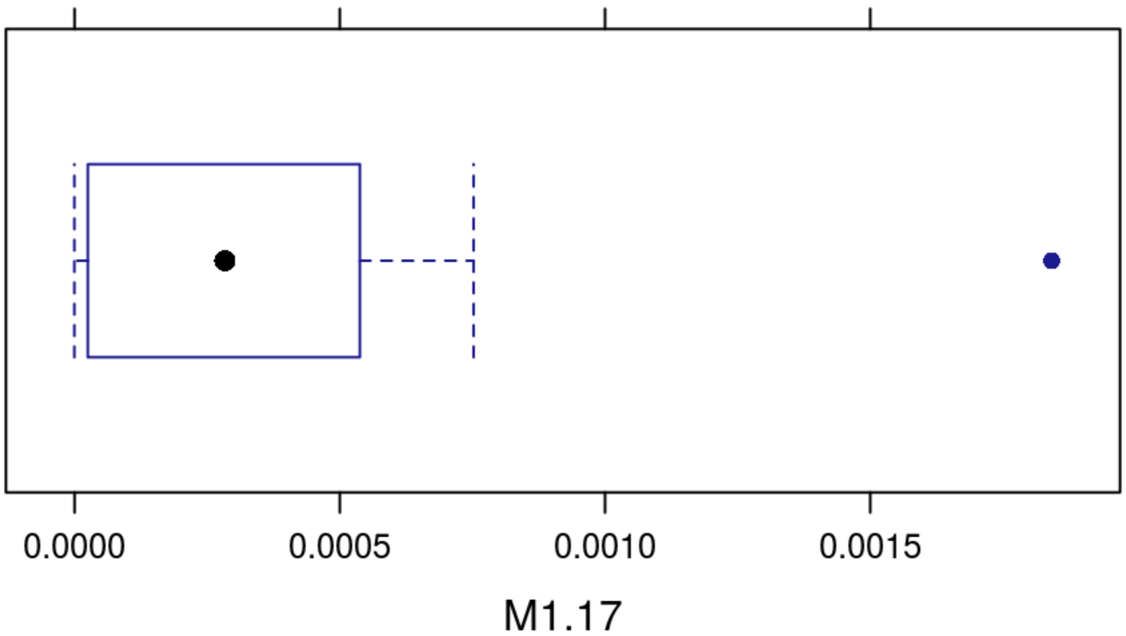

```
bwplot(~M1.15,data=GenderResearch)
```



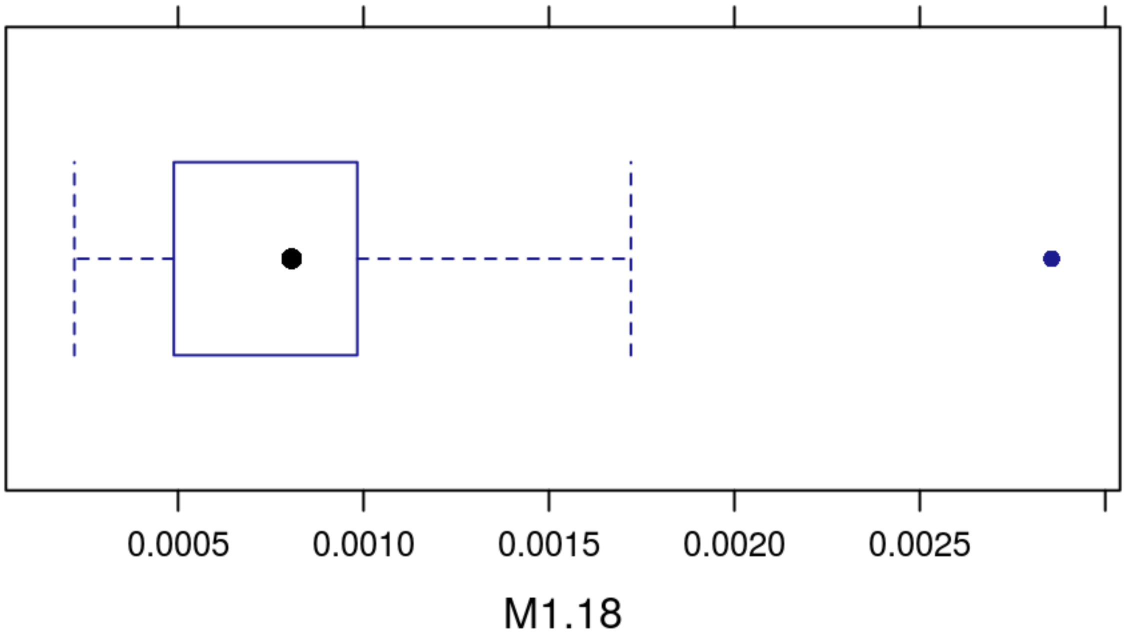

```
bwplot(~M1.16,data=GenderResearch)
```



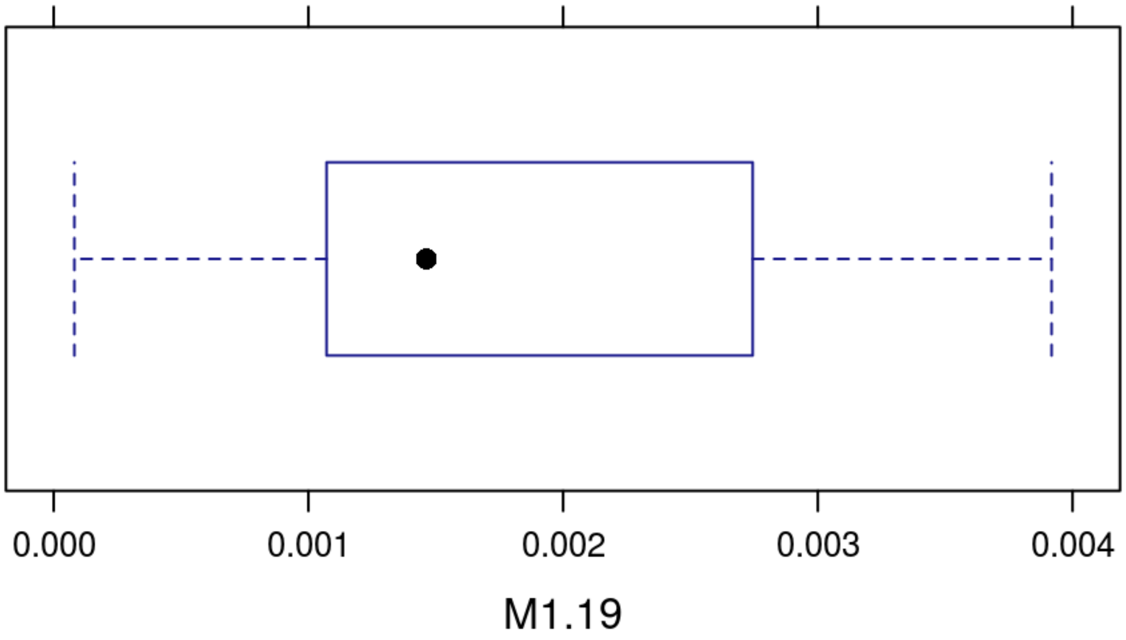

```
bwplot(~M1.17,data=GenderResearch)
```



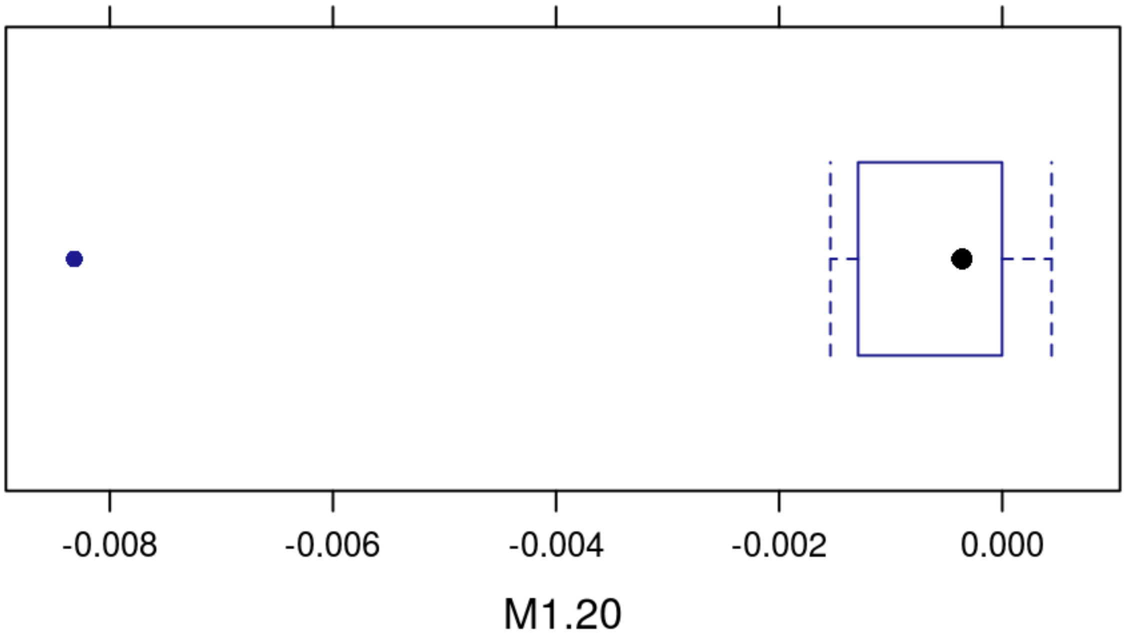

```
bwplot(~M1.18,data=GenderResearch)
```



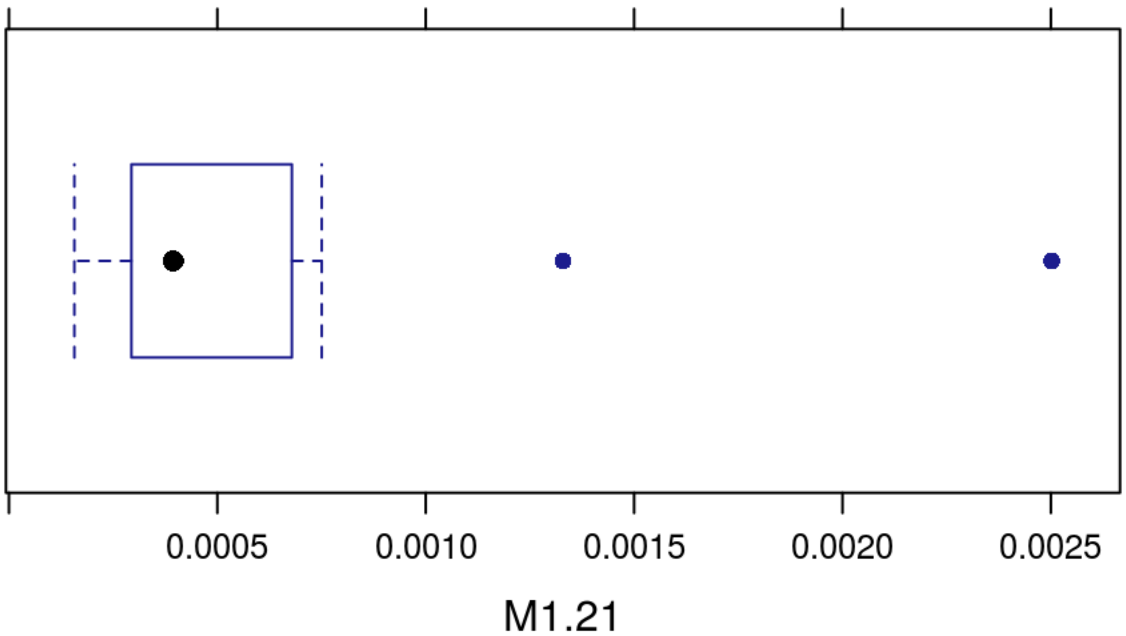

```
bwplot(~M1.19,data=GenderResearch)
```



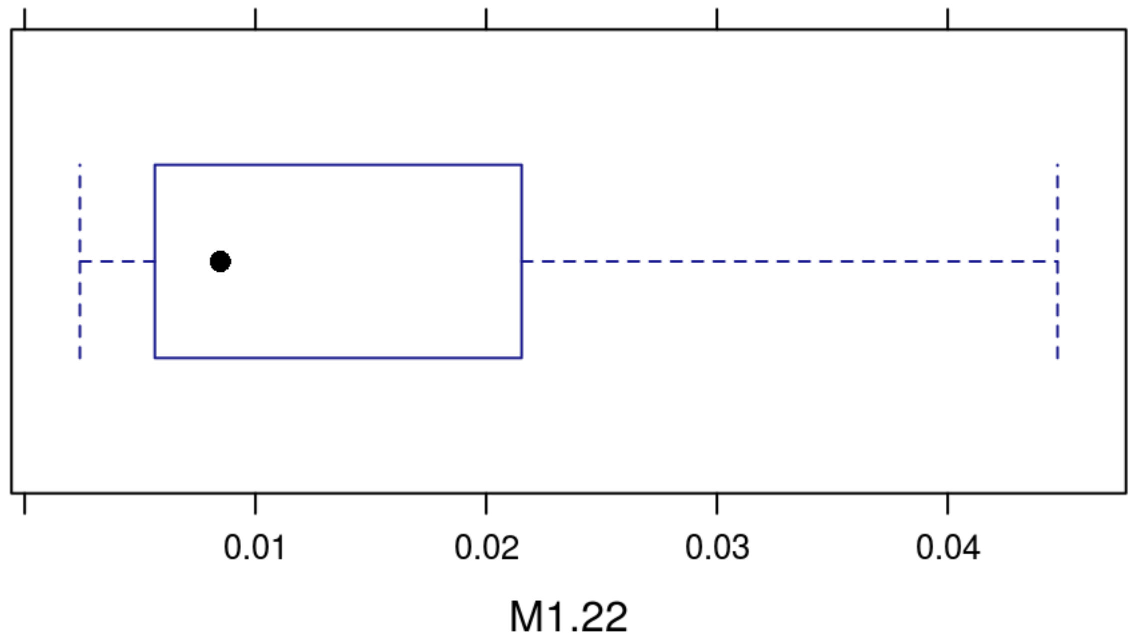

```
bwplot(~M1.20,data=GenderResearch)
```



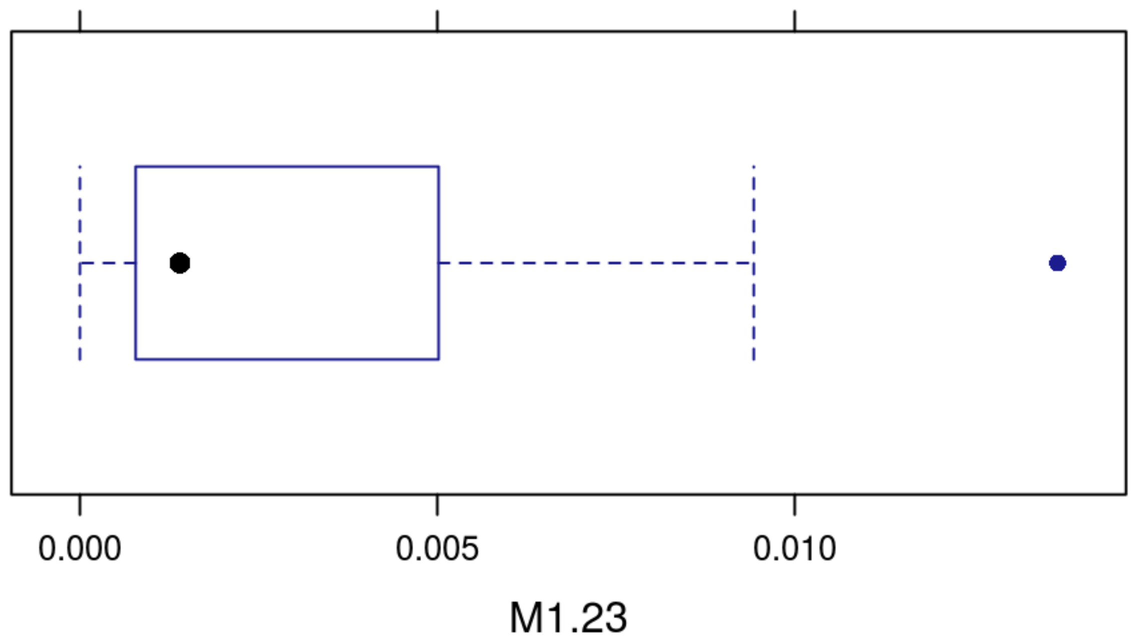

```
bwplot(~M1.21,data=GenderResearch)
```



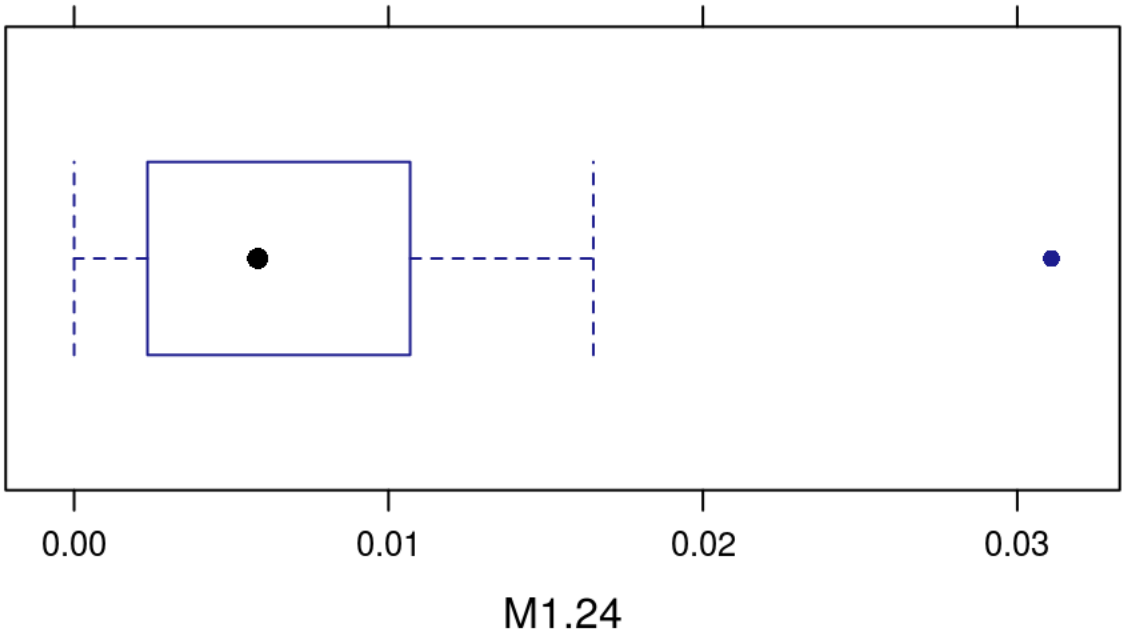

```
bwplot(~M1.22,data=GenderResearch)
```



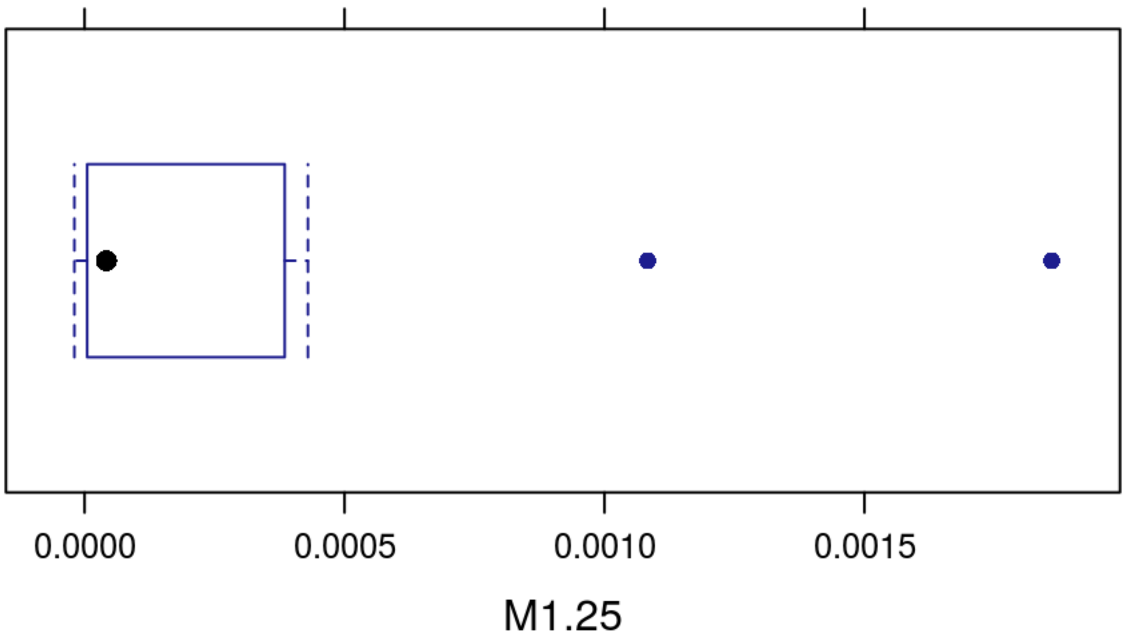

```
bwplot(~M1.23,data=GenderResearch)
```



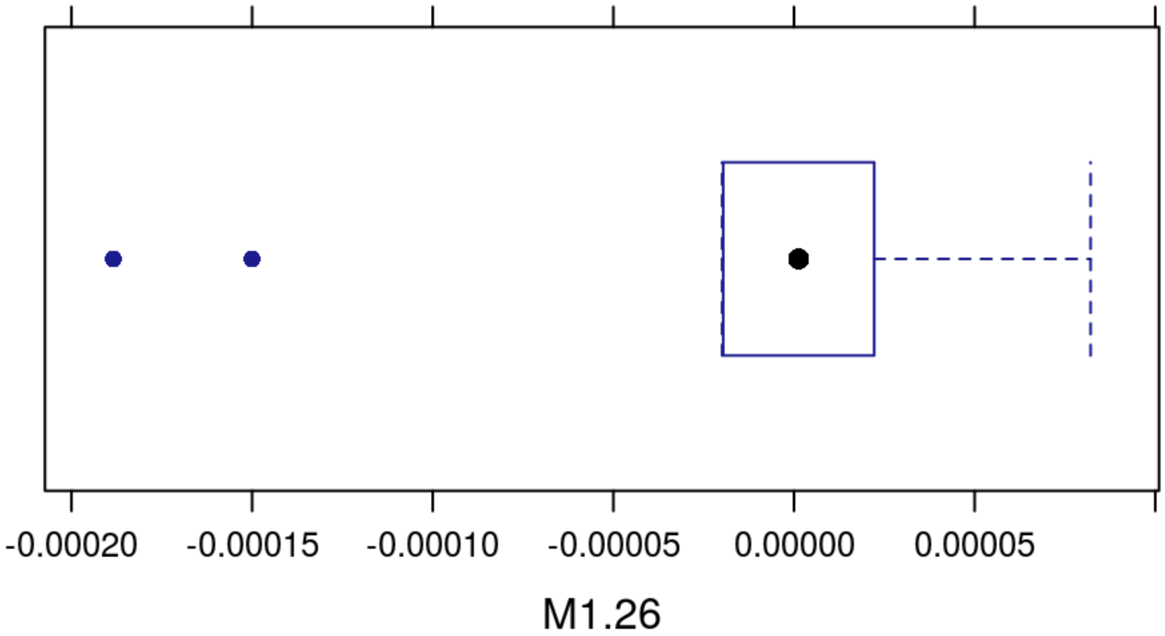

```
bwplot(~M1.24,data=GenderResearch)
```



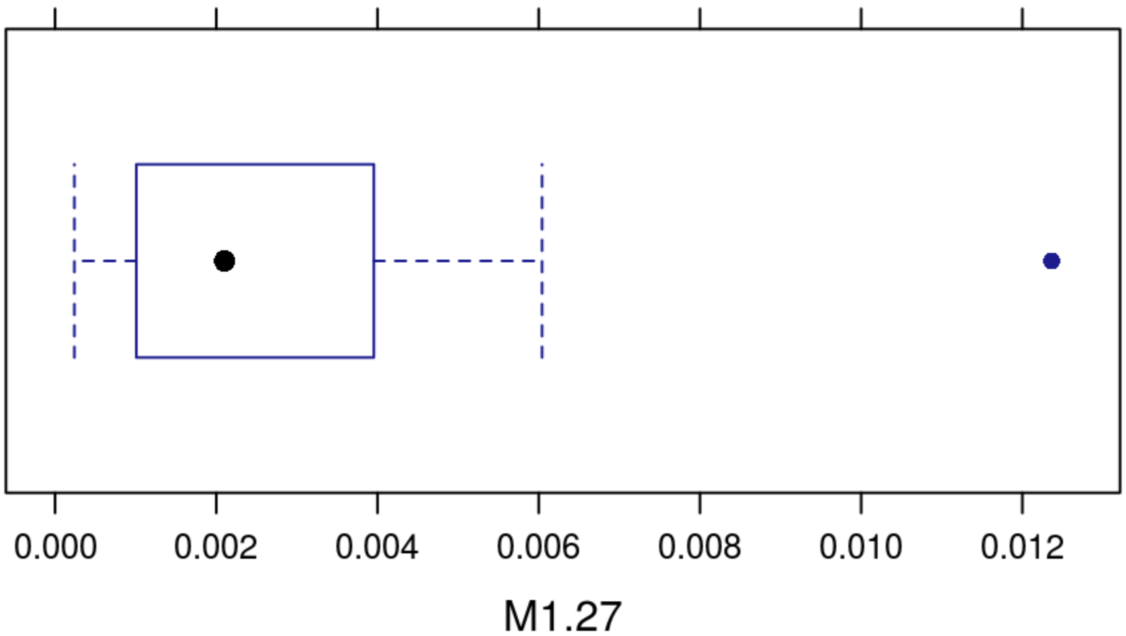

```
bwplot(~M1.25,data=GenderResearch)
```



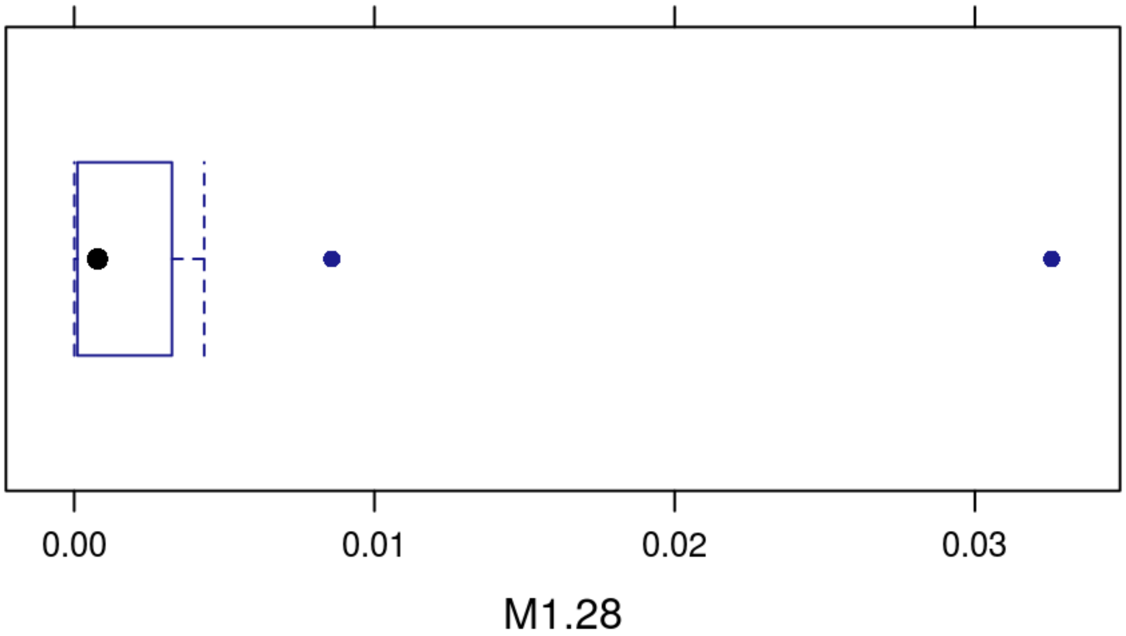

```
bwplot(~M1.26,data=GenderResearch)
```



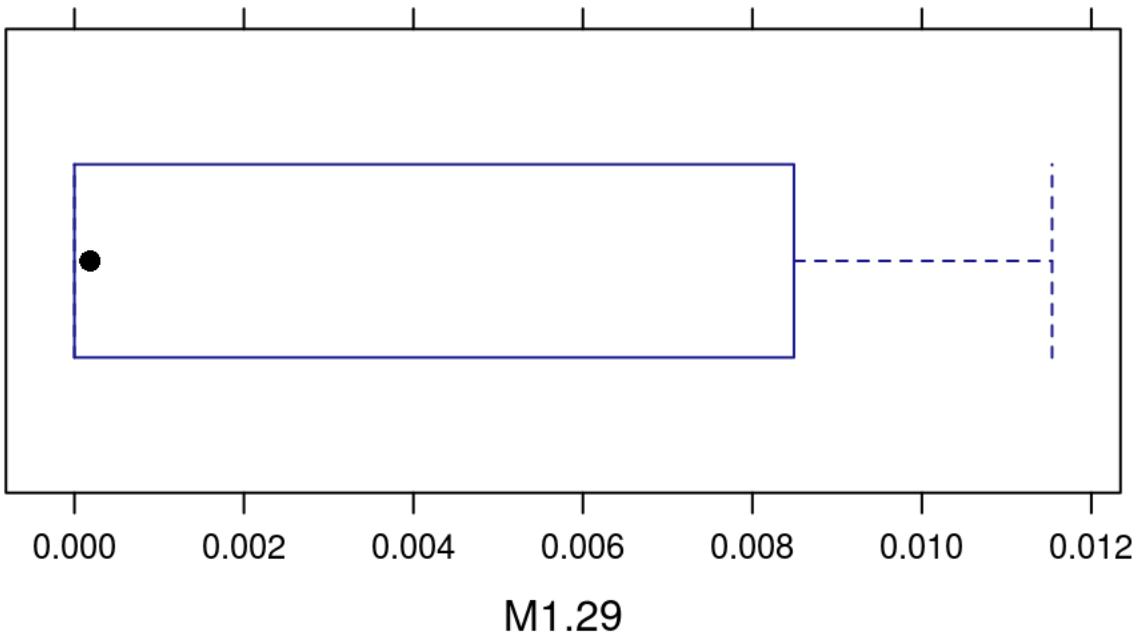

```
bwplot(~M1.27,data=GenderResearch)
```



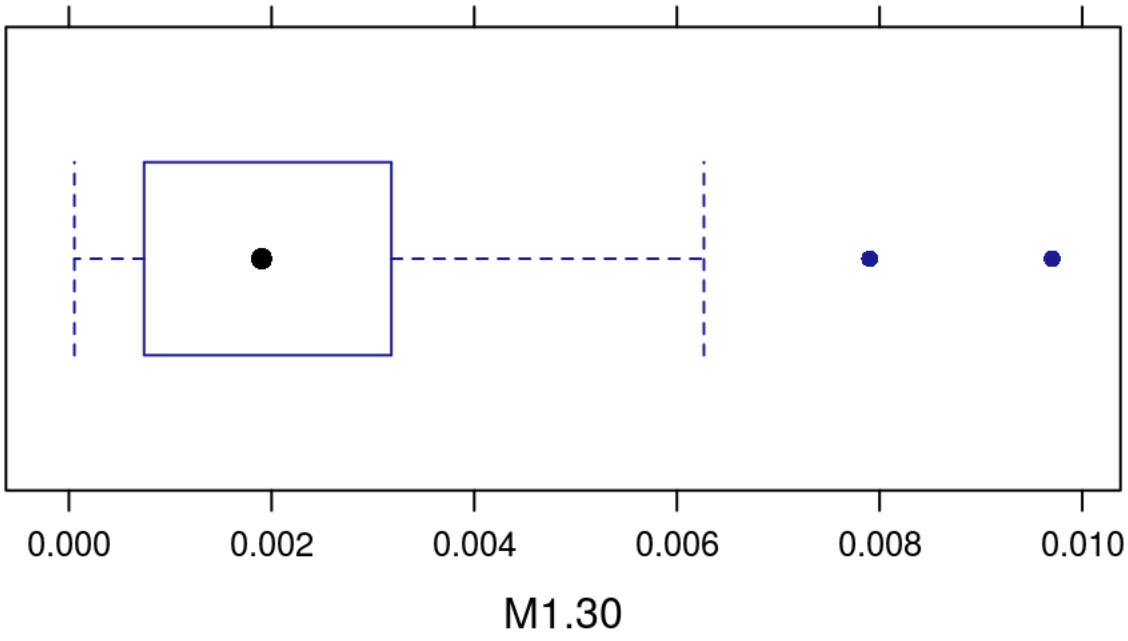

```
bwplot(~M1.28,data=GenderResearch)
```



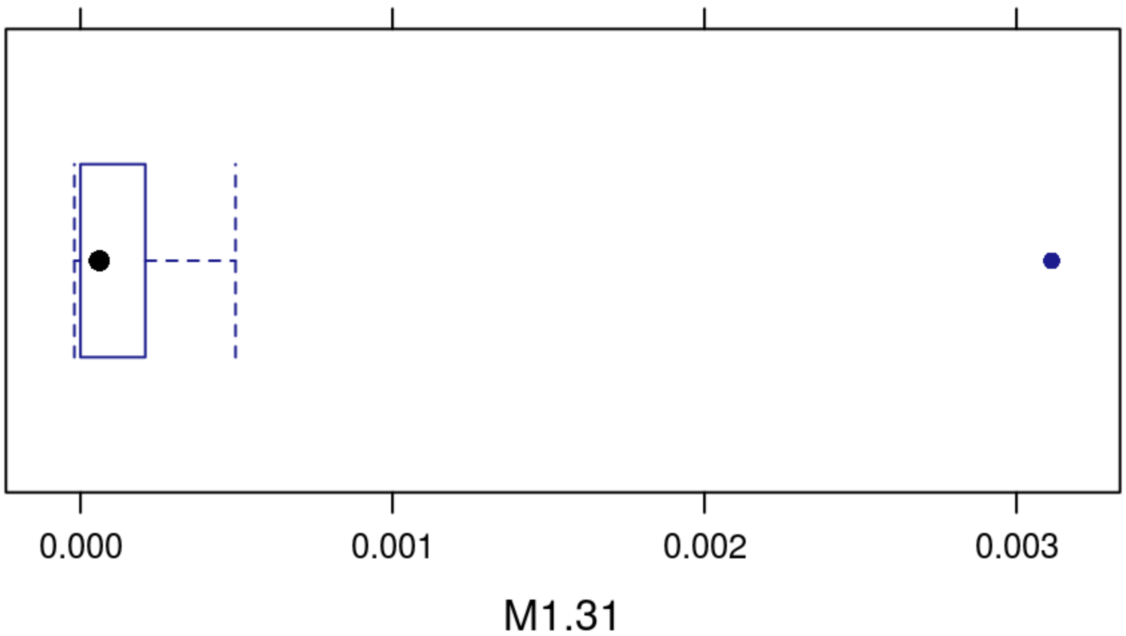

```
bwplot(~M1.29,data=GenderResearch)
```



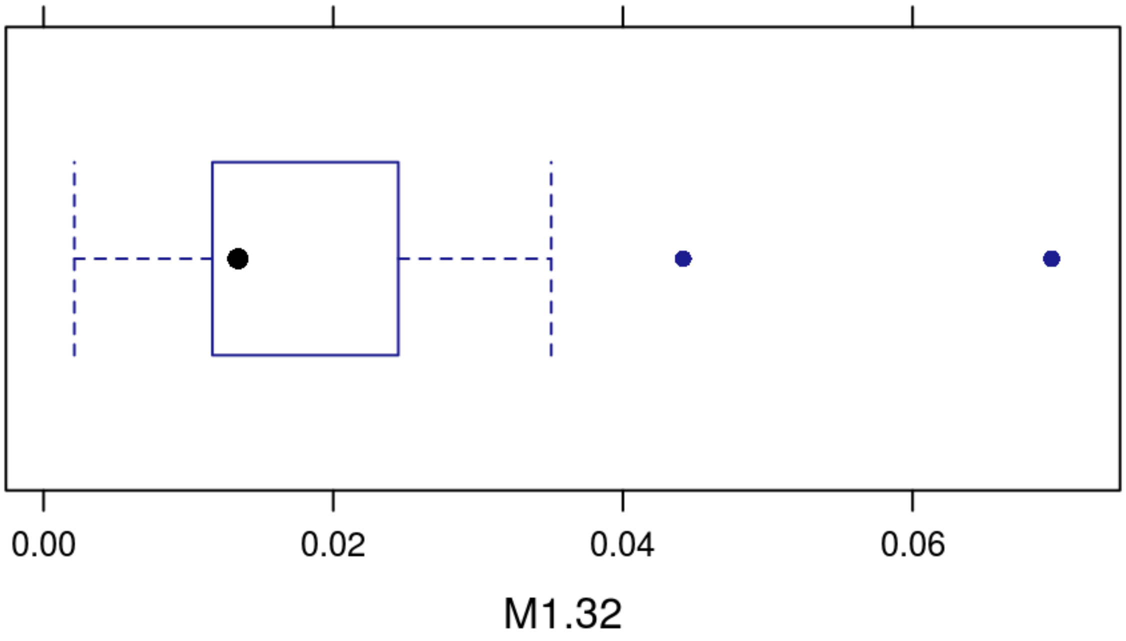

```
bwplot(~M1.30,data=GenderResearch)
```



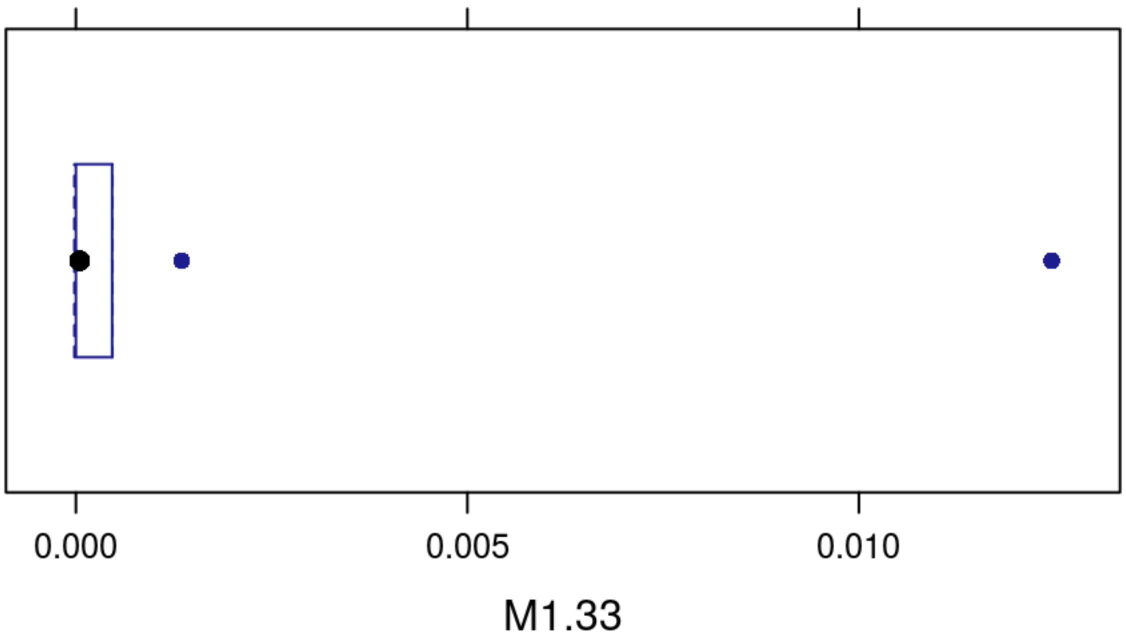

```
bwplot(~M1.31,data=GenderResearch)
```



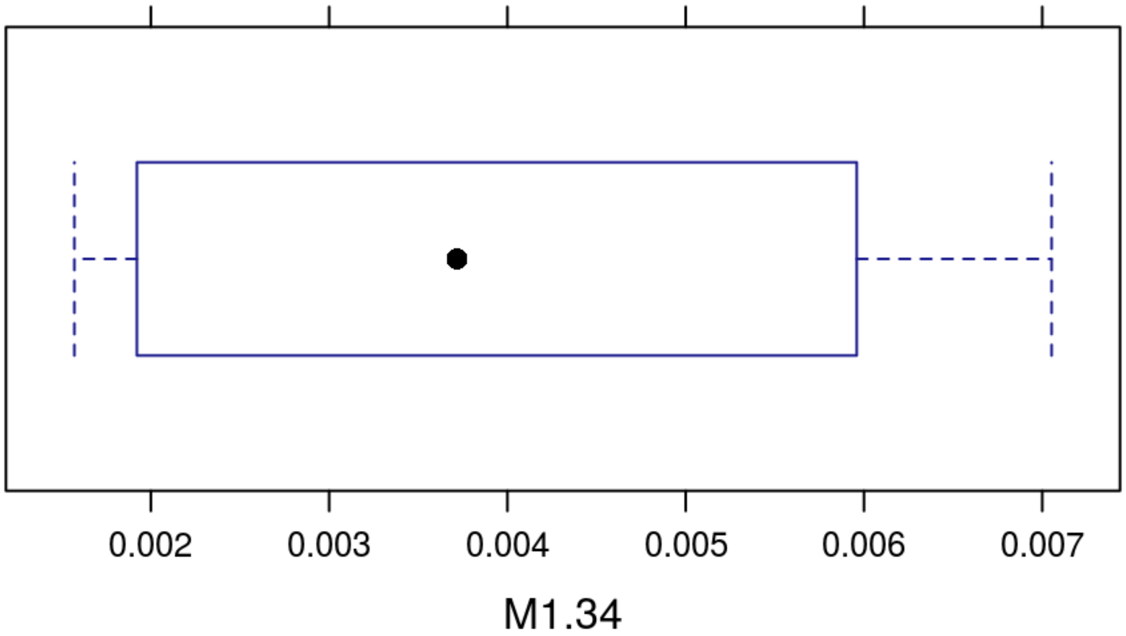

```
bwplot(~M1.32,data=GenderResearch)
```




```
bwplot(~M1.33,data=GenderResearch)
```




```
bwplot(~M1.34,data=GenderResearch)
```




```
bwplot(~M1.35,data=GenderResearch)
```